

Econ 104 Course Project - Applied Econometric Analysis

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Background

Research Question

To what extent does school funding predict student outcomes, and how is that effect moderated by demographic characteristics and/or local tax revenue capacity?

Motivation

We examine whether the level of school funding adequacy affects student outcome gaps across school districts, and whether this relationship depends on the district's property tax capacity. Using panel data from the District Cost Database (DCD) and the Elementary and Secondary Information System (EISi) on approximately 12000 U.S. public school districts across states over a 13 year period, we investigate whether districts with more funding experience smaller student outcome gaps. We also analyze whether the strength of this relationship varies depending on the district's ability to raise revenue locally and the demographics they serve. We hope to uncover similar results to those described in the theory below and better enhance our understanding of the key relationships at play here with regards to specific local and demographic characteristics. The data is somewhat narrow in time but contains a wealth of observations at 132774 so we hope to still achieve meaningful causal estimates. The sampling process is exhaustive with the dataset using government reported data to represent close to all of the public school districts in the United States. Notable exclusions are any district with fewer than 100 students due to common inaccuracies, Hawaii, which has a single government-run school district, and Vermont, which suffers from data irregularities.

Theory

There have been numerous studies on the impact of school funding levels on student outcomes ever since the notorious 1966 Coleman Report, "Equality of Educational Opportunity," argued that school funding and resources were not primary drivers of student achievement. This controversial result has been repeatedly contradicted by more robust economic research of the 21st century. Our current understanding of the relationship between school funding and student outcomes indicates a significant positive relationship moderated by various socioeconomic characteristics. In particular, Jackson and Mackevicus (2024) found that increased school funding improves student outcomes, and through a meta-analysis of 31 studies on the impacts of school spending, were able to directly estimate marginal impacts of funding and evaluate causes of heterogeneity. They found that a sustained \$1,000 increase in per-pupil spending over four years leads to test score increases of 0.0316 standard deviations ($p < 0.001$) and a positive impact more than 90% of the time. They also found that this same increase would lead to a 2% increase in high school graduation and a 2.8% increase in college-going. These effects are not diminishing at current funding levels and are reportedly much stronger on average for economically disadvantaged students compared to advantaged ones (Jackson & Mackevicus, 2024, pp. 414-415). In terms of heterogeneity, we see that the effects of spending are larger for noncapital-specific spending, but not statistically significantly so, and that the marginal dollar spent on capital should be about the same as that spent on labor. Further, there are no statistically significant variations over geographic characteristics such as region or urbanicity (Jackson & Mackevicus, 2024, pp. 438-440).

In the 2022 EPI report "Public education funding in the U.S. needs an overhaul," Allegretto, Garcia, and Weiss outline similar findings regarding positive funding impacts on test scores and other attainment metrics. They note that a sustained 10% increase in per-pupil spending over 12 years led to 0.27 more years of completed education, 7.25% higher wages, and a 3.67% reduction in the annual incidence of adult poverty (2022, pp. 27). They also establish that the majority of education funding comes from state and local sources and that adequacy varies widely across the country, usually being lower on average in higher-poverty districts (2022, pp. 6-10). Further, the effort that states put into education funding varies widely, with many states underinvesting in education relative to capacity (Allegretto et al., 2022, pp. 11-12). Property taxes play a key role in this relationship as a major

revenue source for school districts. On average, students in the poorest districts receive \$5,260 less in local property-tax funding than those in the wealthiest districts (2022, pp. 13-14). The report notably claims that increased education funding can have a countercyclical effect on economic downturns. However, funding typically decreases during these periods, such as in 2008 when education revenues fell sharply and did not recover until about 8 years later, with poorer districts taking even longer (Allegretto et al., 2022, pp. 21).

Hypothesis and Expected Signs

Key variables in our data include:

- **year**: The district year.
- **state**: The state where the district is located.
- **outcomegap**: Gap between district and U.S. average test scores (s.d.) where negative values are below average.
- **fundinggap**: Gap between actual and required spending per pupil.
- **localptax**: Total district revenue from local property taxes.
- **ppcstot**: Actual per-pupil spending totals.
- **predcost**: Required (adequate) spending per pupil.
- **pov**: Census child (5-17 year old) poverty rate.
- **iep**: Percent special education students.
- **ell**: Percent English language learners.
- **enroll**: Total student enrollment.
- **amind**: Percent American Indian/Alaska Native students.
- **asian**: Percent Asian students.
- **black**: Percent Black students.
- **hisp**: Percent Hispanic students.
- **multi**: Percent multiracial students.
- **pac**: Percent Pacific Islander/Hawaii Native students.
- **white**: Percent white students.

Based on the theory discussed above, we expect movement toward funding adequacy reduces student outcome gaps (test scores should increase) and that this effect is stronger in districts with lower property tax capacity or that serve a greater proportion of disadvantaged students. Because schools rely heavily on local property taxes for a sizeable proportion of their funding, differences in property tax capacity may influence a district's ability to generate revenue and respond to funding shortfalls. Districts with stronger tax bases and more advantaged students are better positioned to close funding gaps and invest in resources that reduce disparities in student outcomes. These results should vary systematically across time and across states due to economic shifts and differences in funding policies.

Specifically, we anticipate the following coefficients in a regression on outcomegap (which will increase as outcomes improve):

- $\beta_{fundinggap} > 0$: As funding gaps increase, outcome gaps should do the same (in the same direction, positive or negative).
- $\beta_{localptax} > 0$: As local property tax revenue increases, funding should increase and outcomes should improve.
- $\beta_{ppcstot} > 0$: As the total per pupil spending increases, outcome gaps should improve.
- $\beta_{predcost} < 0$: As the total required spending increases, we expect a higher likelihood of a funding gap or of a concentration of disadvantaged students/decreased resources which could indicate worse outcome gaps.
- $\beta_{pov} < 0$: As the child poverty rate increases, outcome gaps should worsen.

- $\beta_{iep} < 0$: As the proportion of students with special needs increases, adequate funding increases and funding gaps are more likely so outcome gaps worsen.
- $\beta_{ell} < 0$: As the proportion of english language learners increases, adequate funding goes up and funding gaps are more likely so outcome gaps worsen.
- $\beta_{enroll} \leq 0$: As enrollment increases, it is possible that a district is serving a dense urban area and might have resource disparities that lead to slightly worse outcome gaps. We only anticipate a weak negative relationship.
- $\beta_{minority} < 0$: As the proportions of each of the minority ethnicity variables increase, we expect a greater likelihood of funding inadequacy and worse outcome gaps due to historical inequalities.

We also anticipate relationships between our variables with likely interactions between localptax and funding related variables

Analysis

Setup

```
rm(list=ls(all=TRUE))

# random seed for reproducibility
set.seed(22)

# Loading necessary libraries
library(tidyverse)
library(Boruta)
library(lmtest)
library(plm)
library(corrplot)
library(MASS)
library(patchwork)
library(lmtest)
library(car)
library(gplots)
library(tseries)
library(stargazer)
library(cvTools)
library(caret)
library(boot)
library(texreg)
library(broom)
```

```
# loading updated District Cost Database file (with appended property tax column)
dcd <- read.csv("DCD_2025_updated.csv")
cat("Variable Names: ")
```

```
## Variable Names:
```

```
cat(names(dcd), sep = ", ")
```

```
## year, leaid, district, state_name, stabbr, localptax, ppcstot, predcost, fundinggap, outcomegap, enroll, pov, iep, ell, amind, asian, black, hisp, multi, pac, white
```

```
# Basic data exploration/cleaning
colSums(is.na(dcd))
```

```
##      year      leaid  district state_name  stabbr  localptax  ppcstot
##      0         0        0          0        0        11336        0
##  predcost fundinggap outcomegap  enroll    pov      iep      ell
##      0         0      19034      0        0      2677      342
##      amind      asian      black      hisp    multi      pac      white
##      0         0        0        0        0    19357    21870        0
```

```
# Several thousand NAs in outcomegap, localptax, iep, ell, multi, and pac from survey changes over time
```

```
# dropping NA values due to large data size, this will primarily remove observations from 2020-2022,
where outcomegap was not reported
dcd <- drop_na(dcd)
```

```
dim(dcd) # 85697 observations remain
```

```
## [1] 85697    21
```

```
# total minority enrollment variable for simplicity
dcd$minoritysum <- rowSums(dcd[c("amind", "black", "hisp", "pac", "asian")])
```

Variable Selection

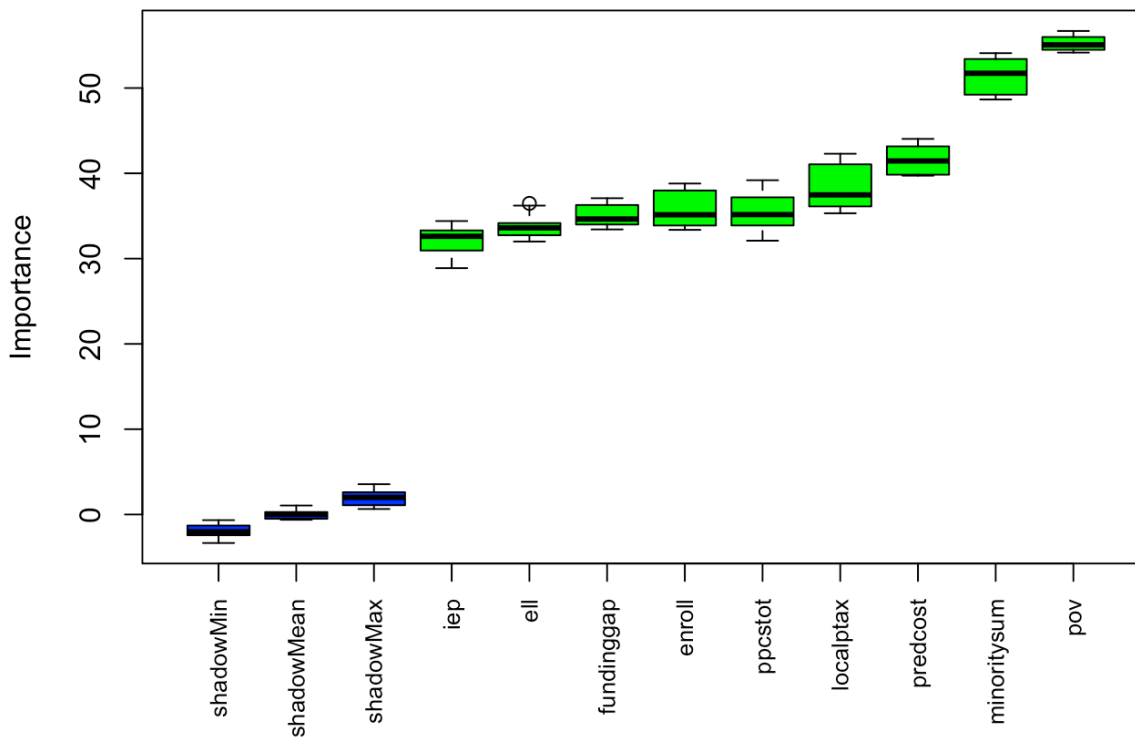
Boruta Algorithm

```
# Boruta Algorithm to identify top quantitative predictors of outcomegap
dcd_sample <- dcd |> slice_sample(n = 10000, replace = FALSE) # sampling to reduce boruta runtime
boruta_out <- Boruta(outcomegap ~ localptax + ppcstot + predcost + fundinggap + enroll + pov + iep +
  ell + minoritysum, data=dcd_sample, doTrace=0) # takes a couple minutes
boruta_out
```

```
## Boruta performed 10 iterations in 1.90059 mins.
## 9 attributes confirmed important: ell, enroll, fundinggap, iep,
## localptax and 4 more;
## No attributes deemed unimportant.
```

```
plot(boruta_out, xlab="", xaxt="n", main="Boruta Feature Importance")
lz <- lapply(1:ncol(boruta_out$ImpHistory), function(i)
  boruta_out$ImpHistory[is.finite(boruta_out$ImpHistory[,i]), i])
names(lz) <- colnames(boruta_out$ImpHistory)
Labels <- sort(sapply(lz, median))
axis(side=1, las=2, labels=names(Labels),
  at=1:ncol(boruta_out$ImpHistory), cex.axis=0.8)
```

Boruta Feature Importance

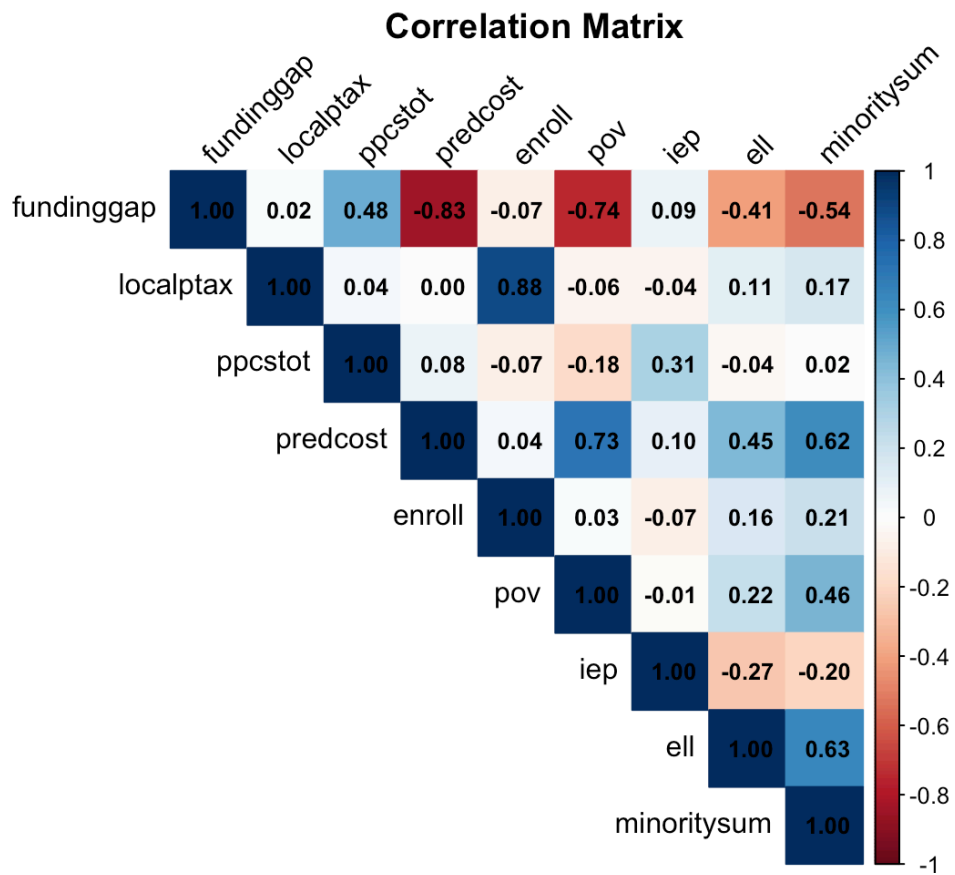


The Boruta Algorithm shows major significance for demographic characteristics like child poverty rate and ethnic composition. We also see spending and property taxes play a significant role in predicting outcomes. All variables are found to be important, though enrollment, fundinggap, ell and iep are the weakest. We might consider dropping them to simplify the model and improve adjusted R^2 .

Correlation Matrix

```
# checking for colinear relationships
corr_matrix <- cor(dcd[c("fundinggap", "localptax", "ppcstot", "predcost", "enroll", "pov", "iep", "ell", "minoritysum")])

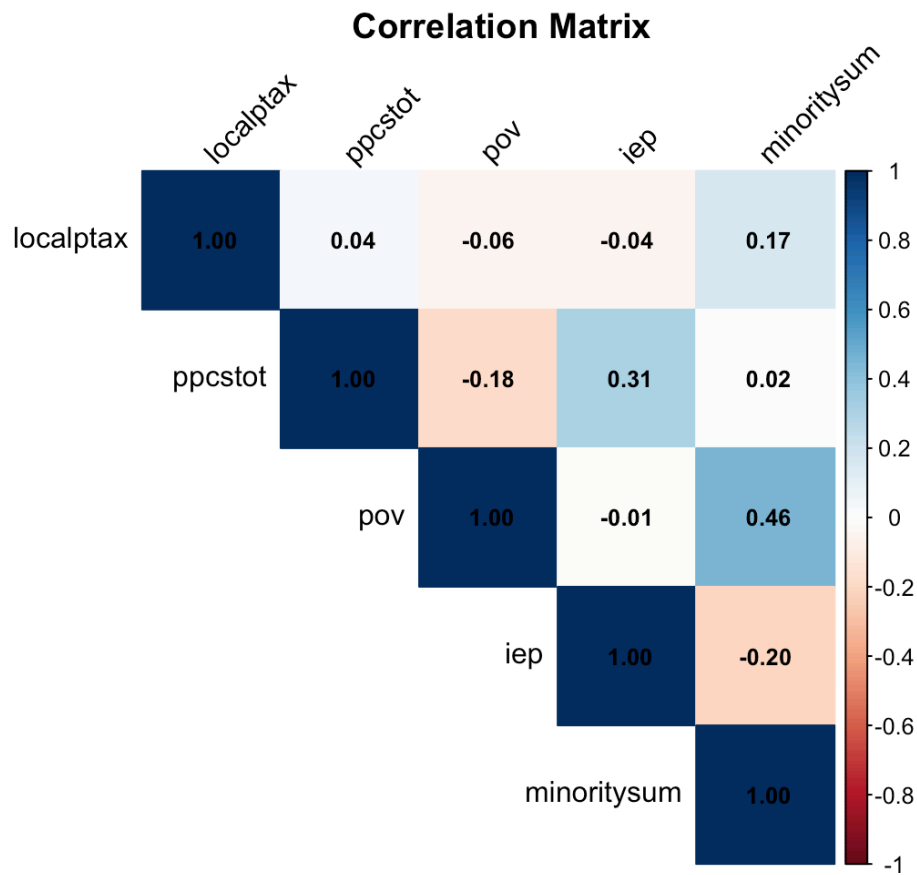
corrplot(corr_matrix, method="color", type="upper",
  addCoef.col="black", number.cex=0.8,
  tl.col="black", tl.srt=45,
  title="Correlation Matrix", mar=c(0,0,1,0))
```



There are strong correlations between enrollment and property taxes, predcost and poverty level, minoritysum and ell, and poverty level and funding gap, and per pupil spending and funding gap. Since enrollment and ell were of relatively lower importance, we will exclude them. Funding gap is of high importance but poverty rate was the top predictor and, combined with per pupil spending, can hopefully capture the same effects. We exclude funding gap. Lastly, predcost is a weaker predictor than poverty level, so we will exclude it.

```
# running again with remaining variables
corr_matrix <- cor(dcd[c("localtax", "ppcstot", "pov", "iep", "minoritysum")])

corrplot(corr_matrix, method="color", type="upper",
  addCoef.col="black", number.cex=0.8,
  tl.col="black", tl.srt=45,
  title="Correlation Matrix", mar=c(0,0,1,0))
```



Correlations look much better with these reduced variables, all values are within ± 0.5 . The highest correlation is between pov and minoritysum, we will keep this in mind when assessing collinearity.

ANOVA Evaluating State and Year Factor Variables

```
## Testing Need for State and Year Effects
m0_pooled <- lm(outcomegap ~ localptax + ppcstot + pov + iep + minoritysum, data = dcd) # pooled
m0_state <- lm(outcomegap ~ localptax + ppcstot + pov + pov + iep + minoritysum + factor(stabbr), data = dcd) # state effects
anova(m0_pooled, m0_state)
```

```
## Analysis of Variance Table
##
## Model 1: outcomegap ~ localptax + ppcstot + pov + iep + minoritysum
## Model 2: outcomegap ~ localptax + ppcstot + pov + pov + iep + minoritysum +
##      factor(stabbr)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1  85691 5386.1
## 2  85651 4747.2 40    638.93 288.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
m0_time <- lm(outcomegap ~ localptax + ppcstot + pov + pov + iep + minoritysum + factor(year), data = dcd) # time effects
anova(m0_pooled, m0_time)
```

```
## Analysis of Variance Table
##
## Model 1: outcomegap ~ localptax + ppcstot + pov + iep + minoritysum
## Model 2: outcomegap ~ localptax + ppcstot + pov + pov + iep + minoritysum +
##      factor(year)
##      Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   85691 5386.1
## 2   85681 5252.3 10      133.82 218.31 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
m0_both <- lm(outcomegap ~ localptax + ppcstot + pov + iep + minoritysum + factor(stabbr) + factor(year), data = dcd) # state and time effects
anova(m0_pooled, m0_both)
```

```
## Analysis of Variance Table
##
## Model 1: outcomegap ~ localptax + ppcstot + pov + iep + minoritysum
## Model 2: outcomegap ~ localptax + ppcstot + pov + iep + minoritysum +
##      factor(stabbr) + factor(year)
##      Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   85691 5386.1
## 2   85641 4636.1 50      750.07 277.12 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(m0_pooled, m0_state, m0_time, m0_both)
```

```
##           df          AIC
## m0_pooled  7  6088.673
## m0_state  47 -4652.440
## m0_time   17  3952.563
## m0_both   57 -6662.611
```

```
BIC(m0_pooled, m0_state, m0_time, m0_both)
```

```
##           df          BIC
## m0_pooled  7  6154.183
## m0_state  47 -4212.587
## m0_time   17  4111.659
## m0_both   57 -6129.173
```

Including both state and year effects improve the model. ANOVA confirms that the new terms are significant and the AIC and BIC are both lowest when we include both state and time effects (with a greater share of the improvement coming from state effects).

Stepwise AIC to Confirm Selection

```
step(m0_both)
```

```
## Start: AIC=-249862.2
## outcomegap ~ localptax + ppcstot + pov + iep + minoritysum +
## factor(stabbr) + factor(year)
##
##           Df Sum of Sq   RSS   AIC
## <none>                4636.1 -249862
## - localptax          1    47.25 4683.3 -248995
## - ppcstot             1    63.10 4699.2 -248706
## - factor(year)       10   111.14 4747.2 -247852
## - iep                 1   234.80 4870.9 -245630
## - minoritysum         1   502.39 5138.5 -241047
## - factor(stabbr)     40   616.24 5252.3 -239247
## - pov                 1  1291.15 5927.2 -228809

##
## Call:
## lm(formula = outcomegap ~ localptax + ppcstot + pov + iep + minoritysum +
## factor(stabbr) + factor(year), data = dcd)
##
## Coefficients:
## (Intercept)          localptax          ppcstot              pov
## 3.147e-01          3.974e-10          1.244e-05          -1.741e+00
## iep          minoritysum factor(stabbr)AR factor(stabbr)AZ
## -1.551e+00          -4.145e-01          3.553e-02          1.080e-01
## factor(stabbr)CA factor(stabbr)DE factor(stabbr)FL factor(stabbr)GA
## 6.403e-03          1.576e-01          2.781e-01          1.486e-01
## factor(stabbr)IA factor(stabbr)ID factor(stabbr)IL factor(stabbr)IN
## 2.795e-03          -8.316e-03          7.109e-02          2.165e-01
## factor(stabbr)KS factor(stabbr)KY factor(stabbr)LA factor(stabbr)ME
## 1.168e-01          2.123e-01          -3.216e-03          9.328e-04
## factor(stabbr)MI factor(stabbr)MN factor(stabbr)MO factor(stabbr)MS
## -5.339e-02          1.385e-01          8.162e-02          1.822e-01
## factor(stabbr)MT factor(stabbr)ND factor(stabbr)NE factor(stabbr)NH
## 6.239e-02          -6.130e-02          2.855e-02          1.500e-01
## factor(stabbr)NJ factor(stabbr)NM factor(stabbr)NV factor(stabbr)NY
## 2.806e-01          1.024e-01          -1.467e-01          -5.815e-02
## factor(stabbr)OH factor(stabbr)OK factor(stabbr)OR factor(stabbr)PA
## 1.762e-01          1.424e-01          -6.751e-02          1.817e-01
## factor(stabbr)RI factor(stabbr)SC factor(stabbr)SD factor(stabbr)TN
## 1.496e-01          6.977e-02          3.174e-02          2.149e-01
## factor(stabbr)TX factor(stabbr)UT factor(stabbr)WA factor(stabbr)WI
## 8.453e-02          9.812e-02          1.539e-02          8.562e-02
## factor(stabbr)WV factor(stabbr)WY factor(year)2010 factor(year)2011
## -1.004e-01          4.243e-02          8.149e-02          1.226e-01
## factor(year)2012 factor(year)2013 factor(year)2014 factor(year)2015
## 1.401e-01          1.599e-01          1.413e-01          1.161e-01
## factor(year)2016 factor(year)2017 factor(year)2018 factor(year)2019
## 1.020e-01          8.621e-02          7.069e-02          3.366e-02
```

Stepwise AIC indicates that all current predictors improve the model (lower the AIC) with localptax being the weakest of the group but still beneficial.

Our final variable selection pre-diagnostics is localptax, ppcstot, pov, iep, minoritysum, year, and state. We specify the initial least squares dummy variable fixed effects model with time and state effects:

$$\text{outcomegap} = \beta_0 + \beta_1 \text{localptax}_{st} + \beta_2 \text{ppcstot}_{st} + \beta_3 \text{pov}_{st} + \beta_4 \text{iep}_{st} + \beta_5 \text{minoritysum}_{st} + \alpha_s + \gamma_t + \varepsilon_{st}$$

Descriptive and Exploratory Data Analysis

Summary Statistics

```
# checking the distribution of key variables
summary(dcd[c("outcomegap", "localptax", "fundinggap", "ppcstot", "predcost", "pov", "iep", "minoritysum")])
```

##	outcomegap	localptax	fundinggap	ppcstot
##	Min. : -2.78564	Min. : 1.000e+03	Min. : -64056.0	Min. : 403
##	1st Qu.: -0.20939	1st Qu.: 1.985e+06	1st Qu.: -2953.0	1st Qu.: 8972
##	Median : 0.01813	Median : 4.824e+06	Median : -16.0	Median : 10288
##	Mean : 0.01349	Mean : 1.746e+07	Mean : -512.7	Mean : 11154
##	3rd Qu.: 0.23353	3rd Qu.: 1.381e+07	3rd Qu.: 2482.0	3rd Qu.: 12359
##	Max. : 1.65754	Max. : 3.074e+09	Max. : 54663.0	Max. : 83114
##	predcost	pov	iep	minoritysum
##	Min. : 3802	Min. : 0.0000	Min. : 0.0000	Min. : 0.00000
##	1st Qu.: 8397	1st Qu.: 0.1063	1st Qu.: 0.1096	1st Qu.: 0.05521
##	Median : 10278	Median : 0.1646	Median : 0.1361	Median : 0.15753
##	Mean : 11666	Mean : 0.1800	Mean : 0.1396	Mean : 0.27247
##	3rd Qu.: 13144	3rd Qu.: 0.2380	3rd Qu.: 0.1659	3rd Qu.: 0.42030
##	Max. : 81925	Max. : 1.0000	Max. : 0.6981	Max. : 1.00000

Here we can immediately see that some variables will need scaling. `localptax` is measured as its raw number and is much larger than any other variables with values in the millions. Similarly we might want to scale down `ppcstot` to be in thousands.

Other than that we can see that on average there is a 0.01349 outcome gap with total per pupil of spending around 11154 slightly below the adequate level of 11666. Although we are not including the variable, it is worth noting that there is an average -512.7 per pupil funding gap.

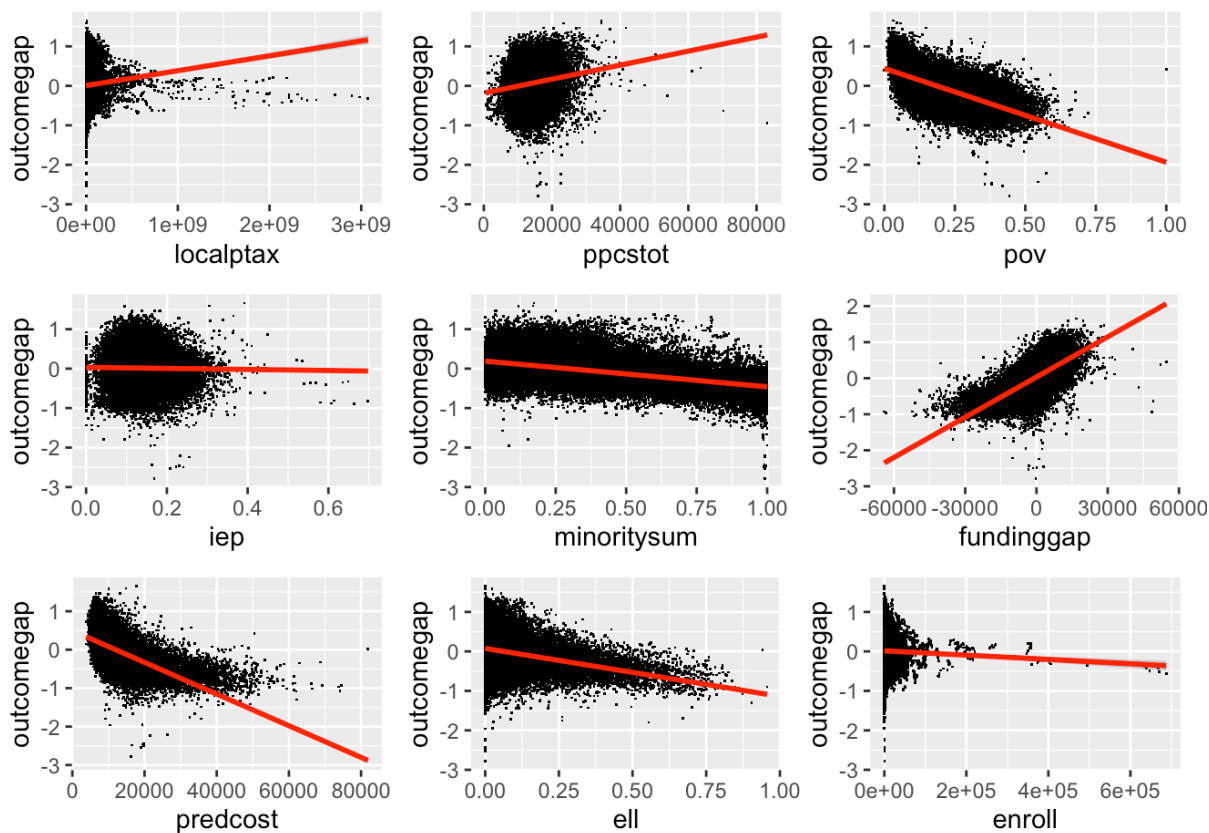
It is also important to note that our gap variables are distributed across 0 with positive values indicating surplus funding and negative values indicating a deficit.

```
# scaling relevant variables
dcd$ppcstot_1000 <- dcd$ppcstot / 1000 # per-pupil spending in 1000s
dcd$localptax_mil <- dcd$localptax / 1e6 # revenue in millions
```

Scatterplots of Outcome Gap Relationships

```
vars <- c("localptax", "ppcstot", "pov", "iep", "minoritysum", "fundinggap", "predcost", "ell", "enroll") # including eliminated variables as well for complete understanding

plots <- list()
for (v in vars){
  p <- ggplot(dcd, aes_string(x=v, y="outcomegap")) +
    geom_point(shape = ".") +
    geom_smooth(method="lm", color="red")
  plots <- append(plots, p)
}
wrap_plots(plots)
```

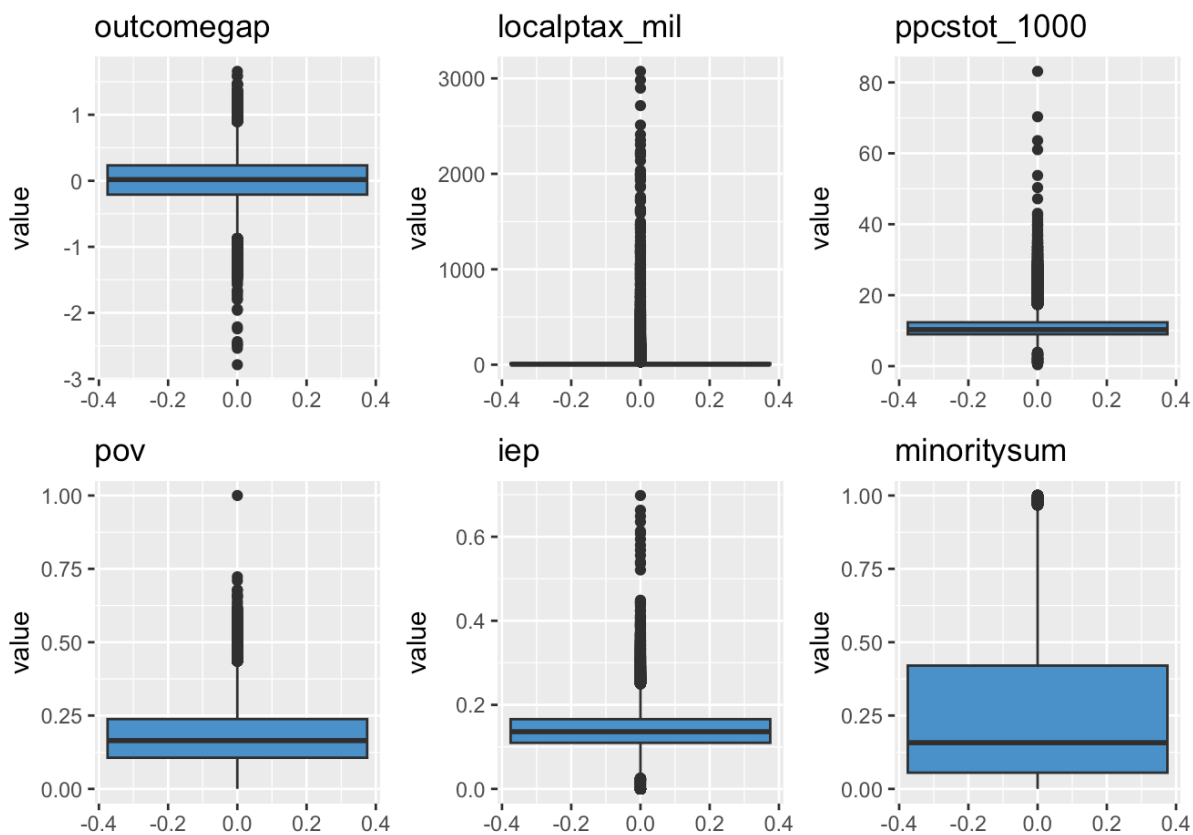


Our predictions for the coefficients of our variables were correct. The trends in outcome gap are as expected. The data does have a right skew for `localptax`, `ppcstot`, `pov`, `predcost`, `ell`, and `enroll` which is to be expected due to the truncation of our dataset to remove the smallest school districts (fewer than 100 students).

Boxplots and Histograms

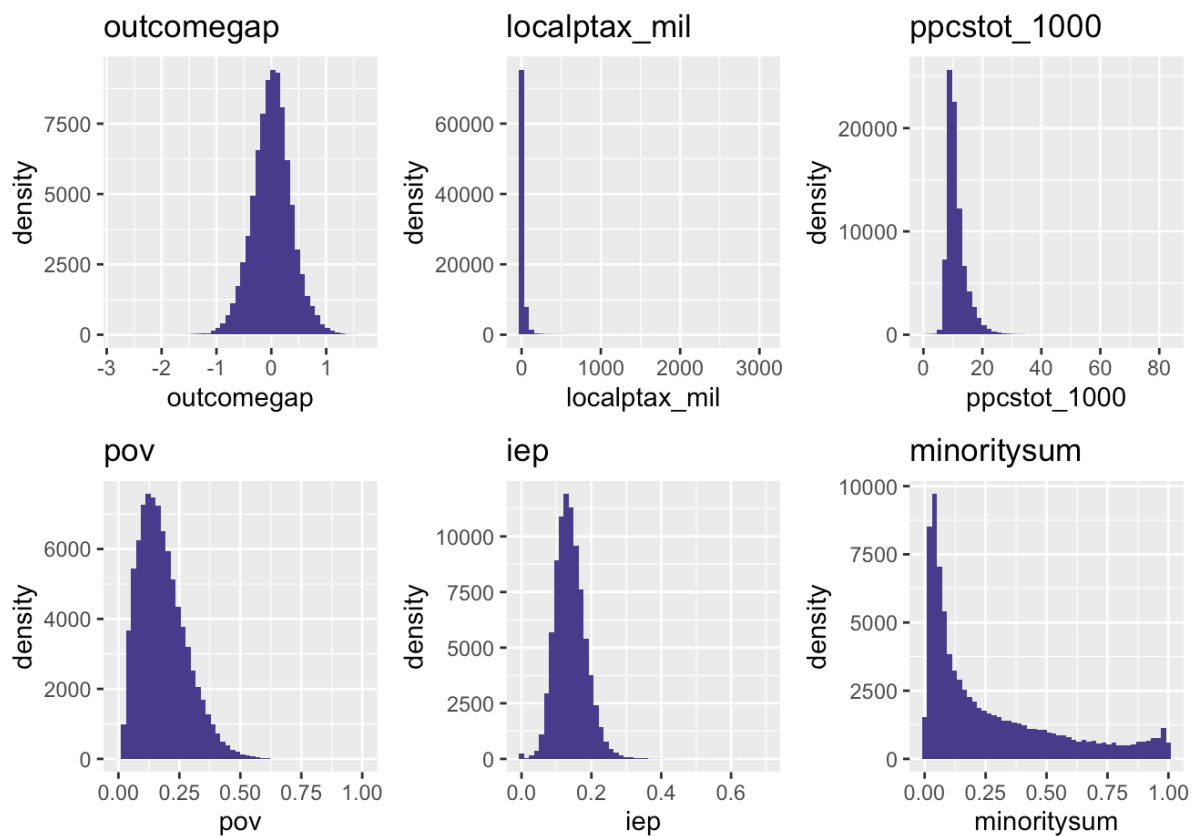
```
vars <- c("outcomegap", "localptax_mil", "ppcstot_1000", "pov", "iep", "minoritysum")

# Boxplots to confirm scaling and observe differences
plots <- list()
for(v in vars){
  p <- ggplot(dcd, aes_string(y=v)) +
    geom_boxplot(fill="steelblue3") +
    labs(y = "value", title = v)
  plots <- append(plots, p)
}
wrap_plots(plots)
```



The scaling looks better, variables are at similar levels. The right skew is particularly evident here and there are some extreme outliers to keep in mind.

```
# Histograms to understand distributions
plots <- list()
for(v in vars){
  p <- ggplot(dcd, aes_string(x=v)) +
    geom_histogram(bins=50, fill="darkslateblue") +
    labs(y = "density", title = v)
  plots <- append(plots, p)
}
wrap_plots(plots)
```



The variables appear to be well distributed though there are high-value outliers and once again we can see the heavy right skew. All in all our skewed variables could benefit from a transformation.

Assessing Trends Across Factor Variables

```
par(mfrow = c(3, 2))

# State
plotmeans(outcomegap ~ stabbr, data = dcd,
  main = "Outcome Gaps by State",
  xlab = "State",
  ylab = "Outcome Gap",
  n.label = FALSE)

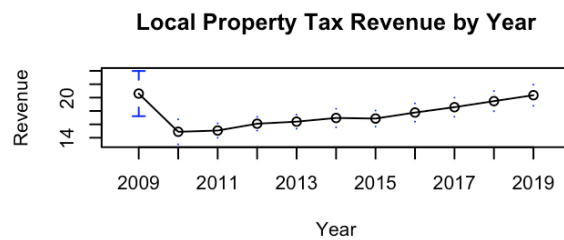
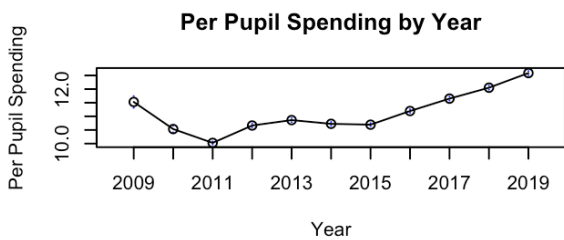
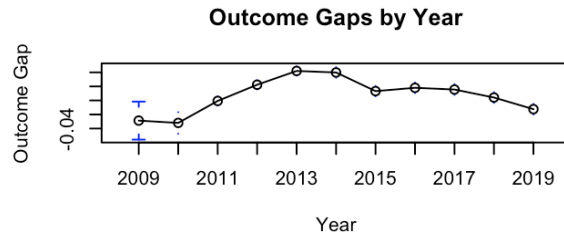
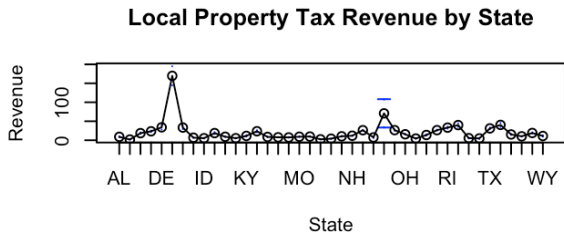
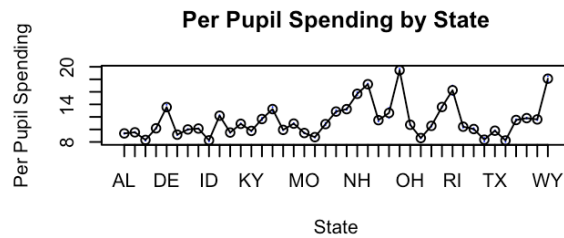
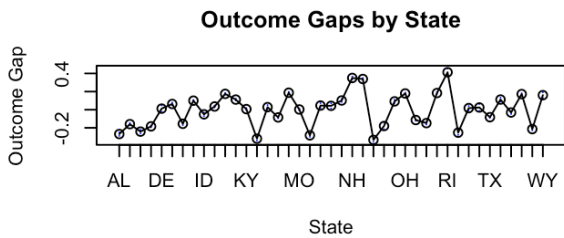
plotmeans(ppcstot_1000 ~ stabbr, data = dcd,
  main = "Per Pupil Spending by State",
  xlab = "State",
  ylab = "Per Pupil Spending",
  n.label = FALSE)

plotmeans(localptax_mil ~ stabbr, data = dcd,
  main = "Local Property Tax Revenue by State",
  xlab = "State",
  ylab = "Revenue",
  n.label = FALSE)

# Year
plotmeans(outcomegap ~ year, data = dcd,
  main = "Outcome Gaps by Year",
  xlab = "Year",
  ylab = "Outcome Gap",
  n.label = FALSE)

plotmeans(ppcstot_1000 ~ year, data = dcd,
  main = "Per Pupil Spending by Year",
  xlab = "Year",
  ylab = "Per Pupil Spending",
  n.label = FALSE)

plotmeans(localptax_mil ~ year, data = dcd,
  main = "Local Property Tax Revenue by Year",
  xlab = "Year",
  ylab = "Revenue",
  n.label = FALSE)
```



```
par(mfrow = c(1,1))
```

Graphs of the mean outcome gap, property tax revenue, and per-pupil spending across factor variables show highly variable state characteristics, as expected. Time trends are consistent with the economic theory suggesting that economic downturns lead to sharp decreases in school funding. In the years following 2008, funding decreases and gaps increase.

Addressing Nonlinear Relationships

Visual inspection of histograms and scatter plots indicates that local property tax revenue, and per-pupil spending are skewed right with many outliers.

Box-Cox Transformation Analysis

```
summary(powerTransform(cbind(dcd$outcomegap, dcd$localptax_mil, dcd$ppcstot_1000, dcd$pov, dcd$iep, dcd$minoritysum) ~ 1, family = "bcnPower")) # box-cox on variables that might require transformations
```

```
## bcnPower transformation to Multinormality
##
## Estimated power, lambda
##      Est Power Rounded Pwr Wald Lwr Bnd Wald Upwr Bnd
## Y1   -0.0168      -0.017   -0.0279   -0.0058
## Y2   -0.0446      -0.045   -0.0483   -0.0409
## Y3   -1.5773      -1.577   -1.6032   -1.5514
## Y4   -1.4286      -1.429   -1.4586   -1.3986
## Y5   -1.3735      -1.374   -1.4254   -1.3216
## Y6   -0.0659      -0.066   -0.0745   -0.0573
##
## Estimated location, gamma
##      Est gamma Std Err. Wald Lower Bound Wald Upper Bound
## Y1      0.7233      NA      NA      NA      NA
## Y2      0.1858      NA      NA      NA      NA
## Y3     11.5129      NA      NA      NA      NA
## Y4      0.4367      NA      NA      NA      NA
## Y5      0.4668      NA      NA      NA      NA
## Y6      0.1000      NA      NA      NA      NA
##
## Likelihood ratio tests about transformation parameters
##                                     LRT df pval
## LR test, lambda = (0 0 0 0 0 0) 27707.58 6    0
## LR test, lambda = (1 1 1 1 1 1) 554253.87 6    0
```

It appears that `outcomegap` and `localptax` both need logarithmic transformations with estimated lambdas close to 0. Meanwhile, our other variables have negative lambdas which indicates the need for an inverse transformation of some sort. This is likely due to the extreme outliers we observed above.

Correcting Outliers

```
vars <- c("outcomegap", "localptax_mil", "ppcstot_1000", "pov", "iep", "minoritysum")
# removing 4 sd outliers (most extreme in our large data) from our key variables directly
# dcd
for (v in vars){
  ub <- mean(dcd[[v]]) + 4 * sd(dcd[[v]])
  lb <- mean(dcd[[v]]) - 4 * sd(dcd[[v]])
  dcd <- filter(dcd, !!as.name(v) > lb & !!as.name(v) < ub)
}
```

For the sake of this analysis, we remove the most extreme outliers. We will not perform the exact box-cox transformations in order to preserve the economic interpretability of our model and we want to ensure our final residuals are close to normally distributed. Inverse transformations of poverty and minority share would affect directional relationships.

Since we are primarily focused on the broader average effects of funding on outcomes rather than the extremes and are data is large, this should not have a major impact on validity. We maintain 84531 observations.

Adjusted Box-Cox

```
summary(powerTransform(cbind(dcd$outcomegap, dcd$localptax_mil, dcd$ppcstot_1000, dcd$pov, dcd$iep, dcd$minoritysum) ~ 1, family = "bcnPower")) # box-cox on variables that might require transformations
(bcn because outcome gap crosses 0)
```

```
## bcnPower transformation to Multinormality
##
## Estimated power, lambda
##      Est Power Rounded Pwr Wald Lwr Bnd Wald Upr Bnd
## Y1   -0.0227      -0.023   -0.0348   -0.0106
## Y2   -0.0142      -0.014   -0.0181   -0.0102
## Y3   -1.3527     -1.353    -1.3795   -1.3259
## Y4   -1.4120     -1.412    -1.4424   -1.3815
## Y5   -0.3847     -0.385    -0.4211   -0.3482
## Y6   -0.0777     -0.078    -0.0864   -0.0691
##
## Estimated location, gamma
##      Est gamma Std Err. Wald Lower Bound Wald Upper Bound
## Y1      0.7821      NA      NA      NA      NA
## Y2      0.1000      NA      NA      NA      NA
## Y3     10.7751      NA      NA      NA      NA
## Y4      0.4345      NA      NA      NA      NA
## Y5      0.2737      NA      NA      NA      NA
## Y6      0.1000      NA      NA      NA      NA
##
## Likelihood ratio tests about transformation parameters
##                                     LRT df pval
## LR test, lambda = (0 0 0 0 0 0) 19221.29 6    0
## LR test, lambda = (1 1 1 1 1 1) 379822.99 6    0
```

The recommended transformations are less extreme than before, notably for IEP rates where extreme outliers shifted the recommended transformation from $\lambda = -1.37$ to -0.38 , and for per-pupil spending which increased from -1.58 to -1.35 . The variables central to our model, local tax revenue and outcome gap, remain stable near zero, confirming the logarithmic transformation. We will test models with these gammas in mind to achieve the best fit that still preserves the interpretability of our model for our research question.

Model Building and Selection

Comparing Transformations

```
# base model
m2_base <- lm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov + iep + minoritysum, data = dcd)
# quadratic demographic terms
m2_quad <- lm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov + I(pov^2) + iep + minoritysum + I(minoritysum^2), data = dcd)
# logarithmic demographic terms
m2_log <- lm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + log(pov + 1) + log(iep + 1) + log(minoritysum + 1), data = dcd)
```

```
anova(m2_base, m2_quad) # Quadratic terms seem to improve the model significantly
```

```
## Analysis of Variance Table
##
## Model 1: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov + iep +
##   minoritysum
## Model 2: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov + I(pov^2) +
##   iep + minoritysum + I(minoritysum^2)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1  84525 4909.5
## 2  84523 4546.3  2    363.12 3375.5 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m2_quad, m2_log) # Log terms offer less improvement
```

```
## Analysis of Variance Table
##
## Model 1: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov + I(pov^2) +
##   iep + minoritysum + I(minoritysum^2)
## Model 2: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + log(pov +
##   1) + log(iep + 1) + log(minoritysum + 1)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1  84523 4546.3
## 2  84525 4869.5 -2   -323.18 3004.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m2_quad) # Quadratic Terms Significant
```

```
##
## Call:
## lm(formula = outcomegap ~ log(localptax_mil) + log(ppcstot_1000) +
##   pov + I(pov^2) + iep + minoritysum + I(minoritysum^2), data = dcd)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.24136 -0.14901 -0.00408  0.14493  1.33477
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.433741   0.008839   49.07  <2e-16 ***
## log(localptax_mil) 0.044918   0.000640   70.18  <2e-16 ***
## log(ppcstot_1000) 0.084552   0.003463   24.41  <2e-16 ***
## pov            -3.830493   0.031231 -122.65  <2e-16 ***
## I(pov^2)         5.502129   0.067833   81.11  <2e-16 ***
## iep             -0.886266   0.020877  -42.45  <2e-16 ***
## minoritysum     -0.241470   0.010496  -23.01  <2e-16 ***
## I(minoritysum^2) -0.319299   0.011711  -27.27  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2319 on 84523 degrees of freedom
## Multiple R-squared:  0.5621, Adjusted R-squared:  0.5621
## F-statistic: 1.55e+04 on 7 and 84523 DF, p-value: < 2.2e-16
```

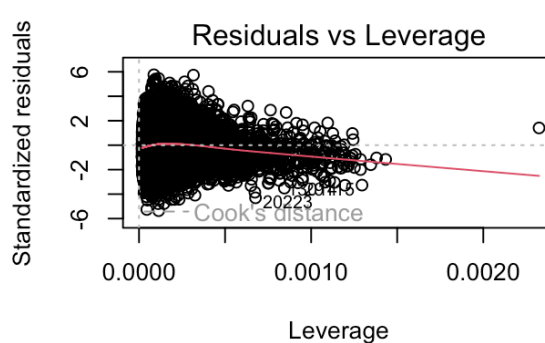
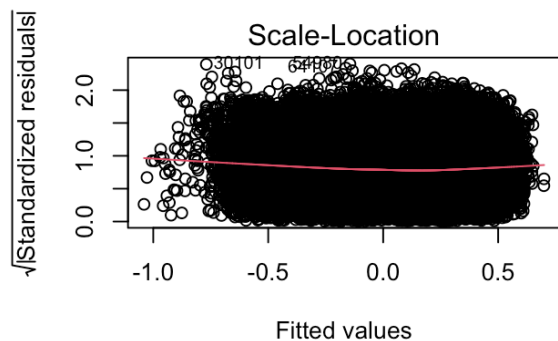
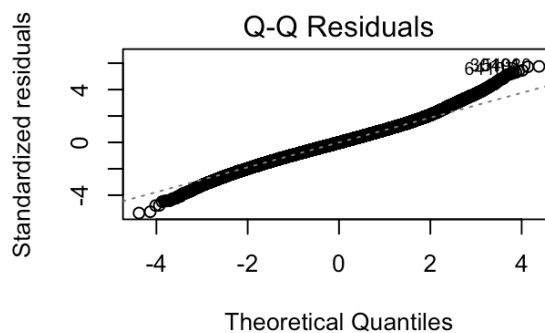
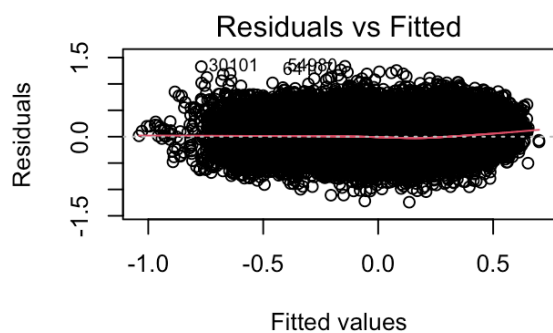
```
AIC(m2_base, m2_quad, m2_log)
```

```
##           df      AIC
## m2_base   7  -668.6352
## m2_quad   9 -7160.1007
## m2_log    7 -1359.0463
```

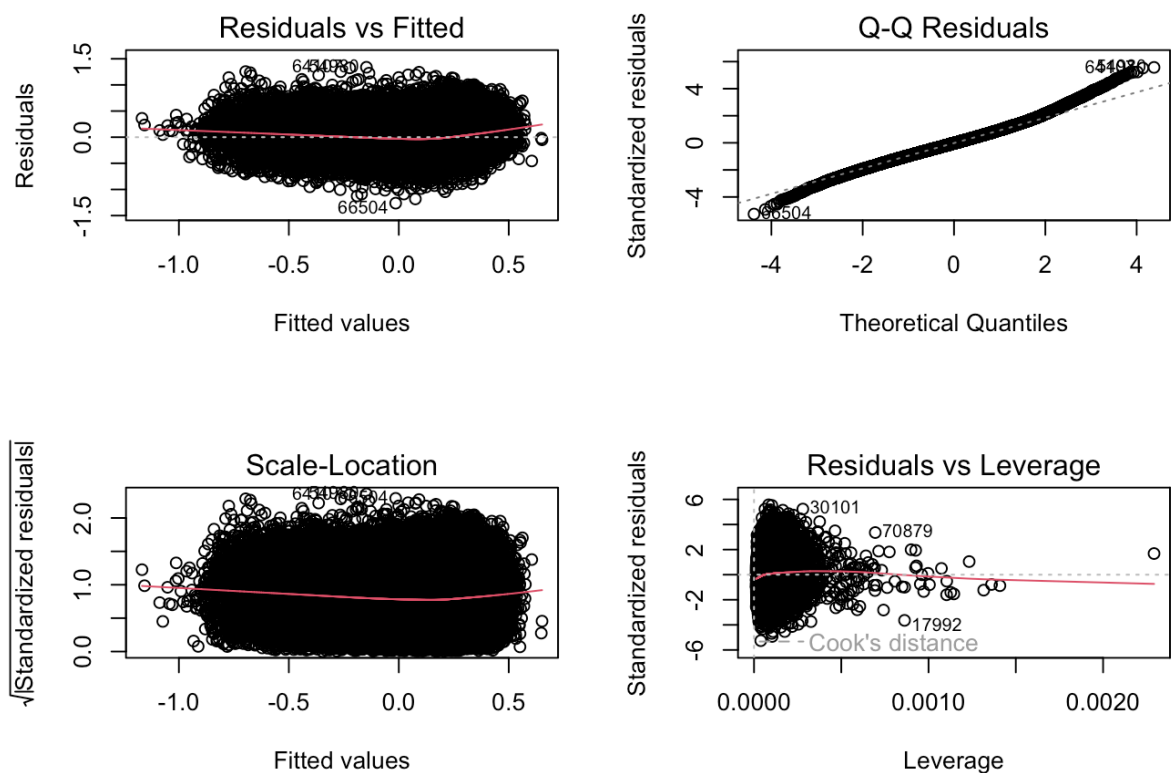
```
BIC(m2_base, m2_quad, m2_log)
```

```
##           df      BIC
## m2_base   7  -603.2211
## m2_quad   9 -7075.9968
## m2_log    7 -1293.6322
```

```
par(mfrow = c(2,2))
plot(m2_quad)
```



```
plot(m2_log)
```



```
par(mfrow = c(1,1))
```

Both logarithmic and quadratic nonlinear terms are significant and improve model fit. Trying to estimate nonlinear terms with a strictly linear model would lead to weak estimation and higher errors. The `m2_quad` model outperforms the others based on AIC and BIC. However, the residuals vs leverage plot shows a slightly stronger downward trend at higher leverage values compared to `m2_log`, which could mean the model is predicting smaller outcome gaps than the true value for more atypical districts. Nonetheless, the diagnostic plots look similar over all and this leverage plot difference is not enough to offset the quadratic model's better fit.

RESET Test

```
resettest(m2_base)
```

```
##
## RESET test
##
## data: m2_base
## RESET = 2649.4, df1 = 2, df2 = 84523, p-value < 2.2e-16
```

```
resettest(m2_quad)
```

```
##
## RESET test
##
## data: m2_quad
## RESET = 1061.8, df1 = 2, df2 = 84521, p-value < 2.2e-16
```

```
resettest(m2_log)
```

```
##  
## RESET test  
##  
## data: m2_log  
## RESET = 2000.5, df1 = 2, df2 = 84523, p-value < 2.2e-16
```

Although we reject the RESET null hypothesis of correct specification for all three models, the quadratic terms are the most significant with the lowest test statistic (1069.8).

VIF

```
vif(m2_quad)
```

```
## log(localptax_mil) log(ppcstot_1000)          pov          I(pov^2)  
##           1.292967           1.260698          14.230398          13.735608  
##                iep        minoritysum  I(minoritysum^2)  
##           1.235386           12.896760          12.880068
```

The Variance Inflation Factor identifies strong colinearity between `pov / minoritysum`, and their quadratic terms. To correct this, we will mean center both variables, reducing multicollinearity while preserving interpretability.

Mean Centering Poverty and Minority Share

```
dcd$pov_c <- scale(dcd$pov, scale = FALSE) # centering to reduce colinearity  
dcd$minority_c <- scale(dcd$minoritysum, scale = FALSE)
```

```
m2_vif <- lm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minorit  
y_c + I(minority_c^2), data = dcd)
```

```
vif(m2_vif)
```

```
## log(localptax_mil) log(ppcstot_1000)          pov_c          I(pov_c^2)  
##           1.292967           1.260698          1.987639          1.468866  
##                iep        minority_c  I(minority_c^2)  
##           1.235386           3.030260          2.722686
```

The VIF looks much better with all parameters below 4. `minority_c` is still highest at 3, which we will keep in mind, but that level does not indicate multicollinearity.

Checking Interaction Terms

```
# defining a model with an interaction between ppcstot and localptax  
m2_no_int <- lm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + pov_c^2 + iep + minorit  
y_c + minority_c^2, data = dcd)  
m2_int <- lm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + log(localptax_mil) * log(ppcstot_1  
000) + pov_c + pov_c^2 + iep + minority_c + minority_c^2, data = dcd)  
  
anova(m2_no_int, m2_int)
```

```
## Analysis of Variance Table
##
## Model 1: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c +
##     pov_c^2 + iep + minority_c + minority_c^2
## Model 2: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + log(localptax_mil) *
##     log(ppcstot_1000) + pov_c + pov_c^2 + iep + minority_c +
##     minority_c^2
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1   84525 4909.5
## 2   84524 4844.0   1    65.486 1142.7 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(m2_no_int, m2_int)
```

```
##           df          AIC
## m2_no_int   7  -668.6352
## m2_int      8 -1801.7520
```

```
BIC(m2_no_int, m2_int)
```

```
##           df          BIC
## m2_no_int   7  -603.2211
## m2_int      8 -1726.9930
```

```
summary(m2_int) # individually significant interaction term
```

```
##
## Call:
## lm(formula = outcomegap ~ log(localptax_mil) + log(ppcstot_1000) +
##     log(localptax_mil) * log(ppcstot_1000) + pov_c + pov_c^2 +
##     iep + minority_c + minority_c^2, data = dcd)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.25902 -0.15530 -0.00545  0.14877  1.34836
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)    0.079488   0.011597    6.854 7.23e-12 ***
## log(localptax_mil) -0.112183   0.004877  -23.000 < 2e-16 ***
## log(ppcstot_1000) -0.001168   0.005057   -0.231   0.817
## pov_c          -1.435703   0.010836 -132.489 < 2e-16 ***
## iep            -1.101776   0.021405  -51.473 < 2e-16 ***
## minority_c      -0.494843   0.003776 -131.057 < 2e-16 ***
## log(localptax_mil):log(ppcstot_1000)  0.069136   0.002045   33.804 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2394 on 84524 degrees of freedom
## Multiple R-squared:  0.5334, Adjusted R-squared:  0.5334
## F-statistic: 1.611e+04 on 6 and 84524 DF,  p-value: < 2.2e-16
```

```
# checking model misspecification
resettest(m2_no_int)
```

```
##
## RESET test
##
## data: m2_no_int
## RESET = 2649.4, df1 = 2, df2 = 84523, p-value < 2.2e-16
```

```
resettest(m2_int) # interaction improved slightly
```

```
##
## RESET test
##
## data: m2_int
## RESET = 2305.3, df1 = 2, df2 = 84522, p-value < 2.2e-16
```

```
vif(m2_int, type="predictor")
```

```
## GVIFs computed for predictors
```

```
##
##          GVIF Df GVIF^(1/(2*Df))    Interacts With
## localptax_mil 151.946365  0          Inf log(ppcstot_1000)
## ppcstot_1000 151.946365  0          Inf log(localptax_mil)
## pov_c        1.607997  1          1.268068          --
## iep          1.218845  1          1.104013          --
## minority_c   1.566415  1          1.251565          --
##
##                                Other Predictors
## localptax_mil localptax_mil, ppcstot_1000, pov_c, iep, minority_c
## ppcstot_1000  localptax_mil, ppcstot_1000, pov_c, iep, minority_c
## pov_c        localptax_mil, ppcstot_1000, iep, minority_c
## iep          localptax_mil, ppcstot_1000, pov_c, minority_c
## minority_c   localptax_mil, ppcstot_1000, pov_c, iep
```

Based on the lower AIC/BIC as well as the improved RESET test and individual significance of the term, an interaction term does improve the model slightly. However, the Variance Inflation Factor indicates that this creates a high level of colinearity so we will forgo the interaction altogether.

We conclude that the following transformations benefit our model fit within the scope of our question:

- Log transformation of right skewed funding variables: `log(localptax_mil)` and `log(ppcstot_1000)`
- Quadratic terms for poverty and minority share: `I(pov_c^2)` and `I(minority_c^2)`
- Mean-centering of poverty and minority share to reduce multicollinearity: `pov_c`, `minority_c`

Competing Model Specifications

Combining all of the above results into final panel data models of our data:

```
# panel data object
dcd.pd <- pdata.frame(dcd, index = c("leaid", "year"))
```

Pooled Model

```
m3_pooled <- plm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minority_c + I(minority_c^2), data = dcd.pd, model = "pooling")
summary(m3_pooled)
```

```
## Pooling Model
##
## Call:
## plm(formula = outcomegap ~ log(localptax_mil) + log(ppcstot_1000) +
##     pov_c + I(pov_c^2) + iep + minority_c + I(minority_c^2),
##     data = dcd.pd, model = "pooling")
##
## Unbalanced Panel: n = 10175, T = 1-11, N = 84531
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -1.241359 -0.149012 -0.004076  0.144934  1.334770
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## (Intercept)   -0.16526373  0.00752455  -21.963 < 2.2e-16 ***
## log(localptax_mil)  0.04491777  0.00064004   70.180 < 2.2e-16 ***
## log(ppcstot_1000)  0.08455182  0.00346317   24.415 < 2.2e-16 ***
## pov_c         -1.85193146  0.01167193 -158.665 < 2.2e-16 ***
## I(pov_c^2)       5.50212889  0.06783324   81.113 < 2.2e-16 ***
## iep           -0.88626598  0.02087734  -42.451 < 2.2e-16 ***
## minority_c     -0.41340014  0.00508776  -81.254 < 2.2e-16 ***
## I(minority_c^2)  -0.31929858  0.01171065  -27.266 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    10382
## Residual Sum of Squares: 4546.3
## R-Squared:      0.5621
## Adj. R-Squared: 0.56206
## F-statistic: 15499.3 on 7 and 84523 DF, p-value: < 2.22e-16
```

Random Effects Model

```
m3_re <- plm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minority_c + I(minority_c^2), data = dcd.pd, model = "random")
```

Fixed Effects Models

```
# One-way (district) fixed effects
m3_ow <- plm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minority_c + I(minority_c^2), model = "within", effect = "individual", data = dcd.pd)

# Two-way fixed effects (district + year)
m3_tw <- plm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minority_c + I(minority_c^2), model = "within", effect = "twoways", data = dcd.pd)
```

```
stargazer(m3_ow, m3_tw, type = "text", title = "Panel Regression 2", column.labels = c("District FE", "Two-Way FE", "Lagged Spending"), dep.var.labels = "Outcome Gap", digits = 3)
```

```
##
## Panel Regression 2
## =====
##                               Dependent variable:
##                               -----
##                               Outcome Gap
##                               District FE      Two-Way FE
##                               (1)              (2)
## -----
## log(localptax_mil)      0.027***            0.034***
##                          (0.003)            (0.003)
##
## log(ppcstot_1000)      0.033***            0.089***
##                          (0.004)            (0.005)
##
## pov_c                   0.133***            -0.081***
##                          (0.017)            (0.017)
##
## I(pov_c2)               0.331***            0.666***
##                          (0.072)            (0.072)
##
## iep                     -0.303***            -0.205***
##                          (0.026)            (0.026)
##
## minority_c              -0.363***            -0.354***
##                          (0.021)            (0.022)
##
## I(minority_c2)          0.050                -0.069*
##                          (0.039)            (0.039)
## -----
## Observations            84,531              84,531
## R2                      0.010                0.012
## Adjusted R2             -0.126              -0.123
## F Statistic             106.140*** (df = 7; 74349) 133.419*** (df = 7; 74339)
## =====
## Note:                    *p<0.1; **p<0.05; ***p<0.01
```

```
# print(pFtest(3_lag, m3_tw))
print(pFtest(m3_tw, m3_ow))
```

```
##
## F test for twoways effects
##
## data:  outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + ...
## F = 228, df1 = 10, df2 = 74339, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

In order to fully account for individual district heterogeneity we substitute the state factor variable for individual district fixed effects. Of the possible fixed effects models, we confirm that both individual and time effects are highly significant ($p < 2.2e-16$), as established with our earlier factor variable estimates. Upon adding year FE, the poverty term sign flips, aligning with our previous estimate for sign (higher poverty → larger gap), meaning that the one way model was impacted by time trends that become isolated with the within transformation. The two-way specification is preferred.

```
summary(m3_tw)
```

```
## Twoways effects Within Model
##
## Call:
## plm(formula = outcomegap ~ log(localptax_mil) + log(ppcstot_1000) +
##      pov_c + I(pov_c^2) + iep + minority_c + I(minority_c^2),
##      data = dcd.pd, effect = "twoways", model = "within")
##
## Unbalanced Panel: n = 10175, T = 1-11, N = 84531
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.88179898 -0.06030317  0.00024998  0.06020259  0.90004238
##
## Coefficients:
##              Estimate Std. Error  t-value Pr(>|t|)
## log(localptax_mil)  0.0335245   0.0029392  11.4060 < 2.2e-16 ***
## log(ppcstot_1000)   0.0891410   0.0052508  16.9768 < 2.2e-16 ***
## pov_c              -0.0812652   0.0174425  -4.6590 3.183e-06 ***
## I(pov_c^2)          0.6655379   0.0718681   9.2605 < 2.2e-16 ***
## iep                -0.2054239   0.0262359  -7.8299 4.948e-15 ***
## minority_c          -0.3535929   0.0221057 -15.9956 < 2.2e-16 ***
## I(minority_c^2)     -0.0689790   0.0391077  -1.7638  0.07777 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    989.71
## Residual Sum of Squares: 977.43
## R-Squared:    0.012407
## Adj. R-Squared: -0.12298
## F-statistic: 133.419 on 7 and 74339 DF, p-value: < 2.22e-16
```

Testing Random vs. Fixed, Pooled vs Fixed

```
phptest(m3_tw, m3_re) #Fixed Effects Preferred
```

```
##
## Hausman Test
##
## data: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + ...
## chisq = 5502, df = 7, p-value < 2.2e-16
## alternative hypothesis: one model is inconsistent
```

```
pFtest(m3_tw, m3_pooled) #Significant Effects compared to pooled
```

```
##
## F test for twoways effects
##
## data: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + ...
## F = 26.653, df1 = 10184, df2 = 74339, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

```
plmtest(m3_pooled, type = "bp") #Individual effects are present
```

```
##
## Lagrange Multiplier Test - (Breusch-Pagan)
##
## data: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + ...
## chisq = 154428, df = 1, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

```
stargazer(m3_pooled, m3_re, m3_tw, type = "text", title = "Panel Regression Results", column.labels =
c("Pooled OLS", "Random Effects", "Fixed Effects District, Year"), dep.var.labels = "Outcome Gap", di
gits = 3)
```

```
##
## Panel Regression Results
## =====
##                               Dependent variable:
##                               -----
##                               Outcome Gap
##                               Random Effects Fixed Effects District, Year
##                               Pooled OLS
##                               (1)           (2)           (3)
## -----
## log(localptax_mil)      0.045***      0.065***      0.034***
##                          (0.001)      (0.001)      (0.003)
##
## log(ppcstot_1000)      0.085***      0.020***      0.089***
##                          (0.003)      (0.004)      (0.005)
##
## pov_c                   -1.852***      -0.429***      -0.081***
##                          (0.012)      (0.015)      (0.017)
##
## I(pov_c2)               5.502***      1.647***      0.666***
##                          (0.068)      (0.069)      (0.072)
##
## iep                    -0.886***      -0.548***      -0.205***
##                          (0.021)      (0.024)      (0.026)
##
## minority_c              -0.413***      -0.583***      -0.354***
##                          (0.005)      (0.010)      (0.022)
##
## I(minority_c2)          -0.319***      -0.205***      -0.069*
##                          (0.012)      (0.024)      (0.039)
##
## Constant                -0.165***      -0.069***
##                          (0.008)      (0.009)
## -----
## Observations            84,531      84,531      84,531
## R2                      0.562      0.132      0.012
## Adjusted R2             0.562      0.132      -0.123
## F Statistic             15,499.280*** (df = 7; 84523) 12,849.310*** 133.419*** (df = 7; 74339)
## =====
## Note:                                     *p<0.1; **p<0.05; ***p<0.01
```

We reject the null hypothesis of the Hausman Test and conclude that two-way fixed effects are preferred, as anticipated given the significance of State effects we already established. The fixed effects model is also confirmed to be a better fit than the pooled model by the Breusch-Pagan test (reject null of no significant individual effects) and the F test for twoways effects.

We can further see that the significance of the poverty term has been reduced slightly, potentially indicating that district effects absorbed some of its impact.

The poverty and IEP coefficients shrink significantly, which suggests that a part of their effect reflects between district differences absorbed by fixed effects rather than within variation.

Checking the Reliability of our Errors

```
jarque.bera.test(residuals(m3_tw))
```

```
##
## Jarque Bera Test
##
## data: residuals(m3_tw)
## X-squared = 33030, df = 2, p-value < 2.2e-16
```

A quick check for normality reveals that our residuals are not normal. We need robust standard errors to rely on any conclusions about parameter significance.

Robust Standard Errors for Model 3

```
coeftest(m3_tw, vcov = vcovHC(m3_tw, type = "HC1", cluster = "group"))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## log(localptax_mil)  0.0335245  0.0052088  6.4361 1.233e-10 ***
## log(ppcstot_1000)   0.0891410  0.0092012  9.6880 < 2.2e-16 ***
## pov_c               -0.0812652  0.0298116 -2.7260 0.006413 **
## I(pov_c^2)           0.6655379  0.1202720  5.5336 3.148e-08 ***
## iep                 -0.2054239  0.0468112 -4.3883 1.144e-05 ***
## minority_c          -0.3535929  0.0424528 -8.3291 < 2.2e-16 ***
## I(minority_c^2)     -0.0689790  0.0730975 -0.9437 0.345348
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The robust estimates tell us that the quadratic term for proportion minority students is insignificant in our model while the poverty term remains significant, we drop `minority_c^2`.

Model 4: Two Way Within Estimator w/o Minority^2 Term

```
m4_fe <- plm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minorit
y_c, data = dcd.pd, model = "within", effect = "twoways")

summary(m4_fe)
```

```
## Twoways effects Within Model
##
## Call:
## plm(formula = outcomegap ~ log(localptax_mil) + log(ppcstot_1000) +
##      pov_c + I(pov_c^2) + iep + minority_c, data = dcd.pd, effect = "twoways",
##      model = "within")
##
## Unbalanced Panel: n = 10175, T = 1-11, N = 84531
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.88169251 -0.06035605  0.00025029  0.06019751  0.90137216
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## log(localptax_mil)  0.0333217  0.0029370  11.3456 < 2.2e-16 ***
## log(ppcstot_1000)   0.0885683  0.0052408  16.8998 < 2.2e-16 ***
## pov_c              -0.0810162  0.0174422  -4.6448 3.409e-06 ***
## I(pov_c^2)          0.6721441  0.0717715   9.3651 < 2.2e-16 ***
## iep                -0.2062173  0.0262325  -7.8611 3.857e-15 ***
## minority_c         -0.3743957  0.0186968 -20.0246 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    989.71
## Residual Sum of Squares: 977.47
## R-Squared:      0.012366
## Adj. R-Squared: -0.12301
## F-statistic: 155.133 on 6 and 74340 DF, p-value: < 2.22e-16
```

The summary of this model indicates that our panel is unbalanced, with districts being observed between 1 and 11 years. This is likely to to our initial outlier removal as well as data differences over the years. This is no issue, as the two way within estimator accomodates unbalanced panels.

Our final model is the two-way within estimator fixed effects model:

$$\text{outcomegap}_{it} = \beta_0 + \beta_1 \log(\text{localptax}_{it}) + \beta_2 \log(\text{ppcstot}_{it}) + \beta_4 \text{pov}_{it} + \beta_5 \text{pov}_{it}^2 + \beta_6 \text{iep}_{it} + \beta_7 \text{minority}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Final Model Diagnostics and Robustness

Re-specifying de-meaned model with `lm()` to work with key diagnostic functions:

```
# convert m4_fe to a lm() since vif(), cooks.distance(), influence(), etc. do not work on plm objects
m4_demeaned_X <- model.matrix(m4_fe) # extracting the de-meaned regressors
m4_demeaned_y <- pmodel.response((m4_fe)) # extracting de-meaned y

# lm object approximation with plm demeaned values and no intercept
m4_lm <- lm(m4_demeaned_y ~ 0 + `log(localptax_mil)` + `log(ppcstot_1000)` + `pov_c` + `I(pov_c^2)` +
`iep` + `minority_c`, data = as.data.frame(m4_demeaned_X))
```

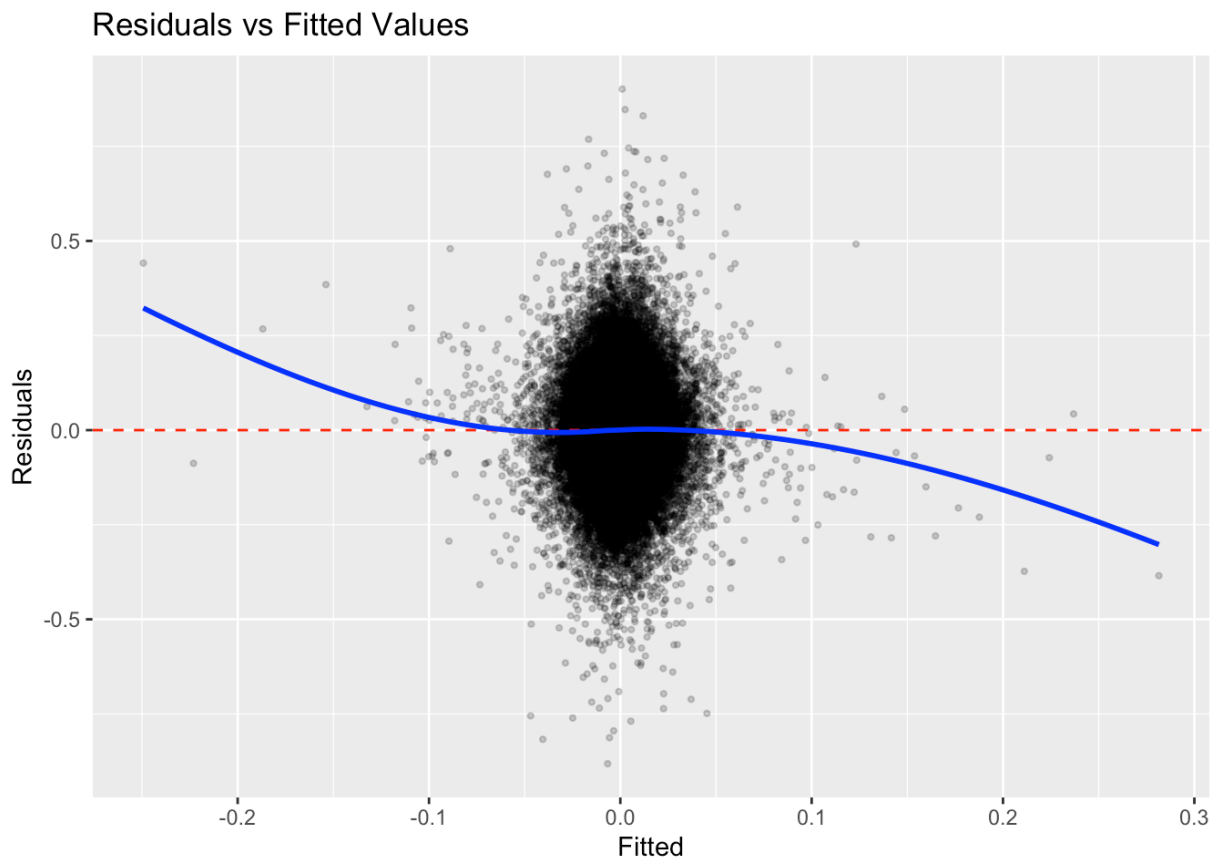
Linearity

Fitted vs Residual Plot

```
# extracting fitted and residual values
m4_data <- data.frame(
  fitted = fitted(m4_fe),
  residuals = residuals(m4_fe))

# fitted vs residuals plot for m4_fe
ggplot(m4_data, aes(x = fitted, y = residuals)) +
  geom_point(alpha = 0.2, size = 0.8) +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  labs(title = "Residuals vs Fitted Values", x = "Fitted", y = "Residuals")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Our model overestimates for districts with low fitted values and underestimates for districts with high fitted values. The bulk of the residuals appear to be randomly distributed about 0, however.

Variance Inflation Factor and Breusch-Godfrey Test

```
#Testing VIF
vif(m4_lm)
```

```
## Warning in vif.default(m4_lm): No intercept: vifs may not be sensible.
```

```
## `log(localptax_mil)` `log(ppcstot_1000)` pov_c
## 1.008766 1.052236 1.443708
## `I(pov_c^2)` iep minority_c
## 1.442106 1.043333 1.002929
```

```
pbgttest(m4_fe)      # Breusch-Godfrey for panels
```

```
##  
## Breusch-Godfrey/Wooldridge test for serial correlation in panel models  
##  
## data: outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + ...  
## chisq = 9974.2, df = 1, p-value < 2.2e-16  
## alternative hypothesis: serial correlation in idiosyncratic errors
```

The Variance Inflation Factor looks good with all model parameters at around 1, well below 4 indicating no significant multicollinearity. The Breusch-Godfrey Test tells us that there is serial correlation in our panel model. We reject the null that errors are uncorrelated with ($p < 2.2e-16$). This indicates a need for cluster robust standard errors.

Homoskedasticity

Breusche-Pagan Test

```
# breusch-pagan  
bptest(m4_fe)
```

```
##  
## studentized Breusch-Pagan test  
##  
## data: m4_fe  
## BP = 3575, df = 6, p-value < 2.2e-16
```

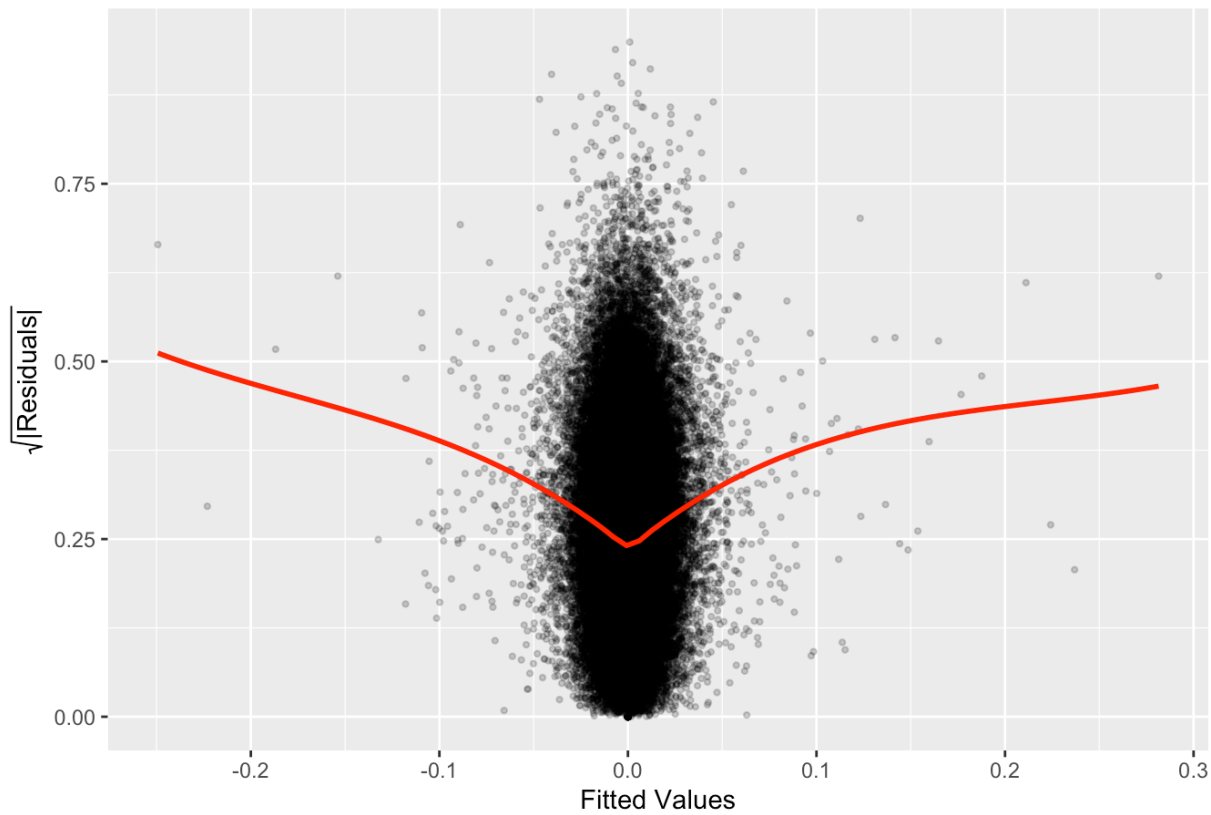
We reject the null of the BP test that errors are homoskedastic ($p < 2.2e-16$). Heteroskedasticity is detected and cluster robust SEs are necessary.

Scale Location Plot

```
m4_data$sqrt_resid <- sqrt(abs(m4_data$residuals))  
  
# Scale-Location  
ggplot(m4_data, aes(x = fitted, y = sqrt_resid)) +  
  geom_point(alpha = 0.2, size = 0.8) +  
  geom_smooth(method = "loess", se = FALSE, color = "red") +  
  labs(title = "Scale-Location", x = "Fitted Values", y = expression(sqrt("|Residuals|")))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Scale-Location



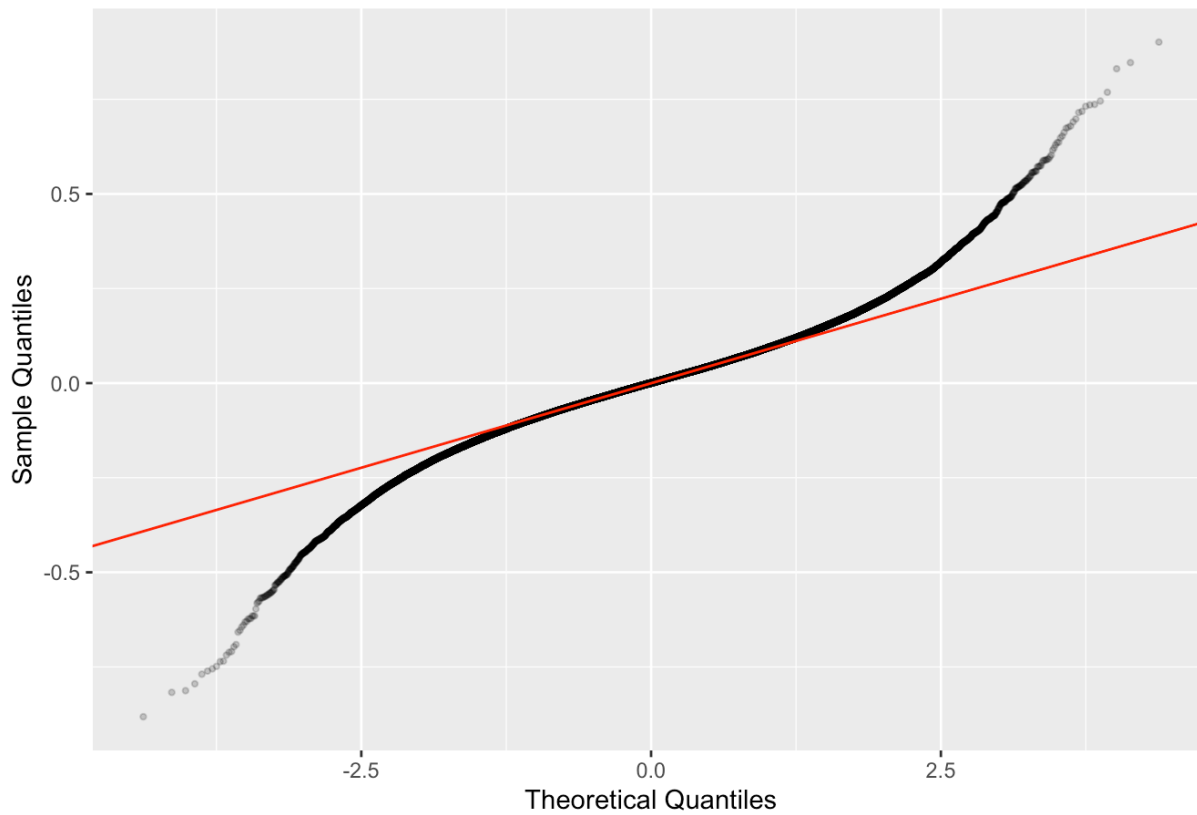
The scale location plot shows a V-shape caused by the within transformation and demeaning rather than heteroskedasticity, but variance is stable across the center of observations.

Normality of Residuals

Q-Q Plot

```
# Q-Q Plot m4
ggplot(m4_data, aes(sample = residuals)) +
  stat_qq(alpha = 0.2, size = 0.8) +
  stat_qq_line(color = "red") +
  labs(title = "Normal Q-Q Plot", x = "Theoretical Quantiles", y = "Sample Quantiles")
```

Normal Q-Q Plot



The Q-Q plot again confirms the heavy tails, consistent with the residuals vs. fitted plot interpretation.

Jarque-Bera Test

```
jarque.bera.test(m4_data$residuals)
```

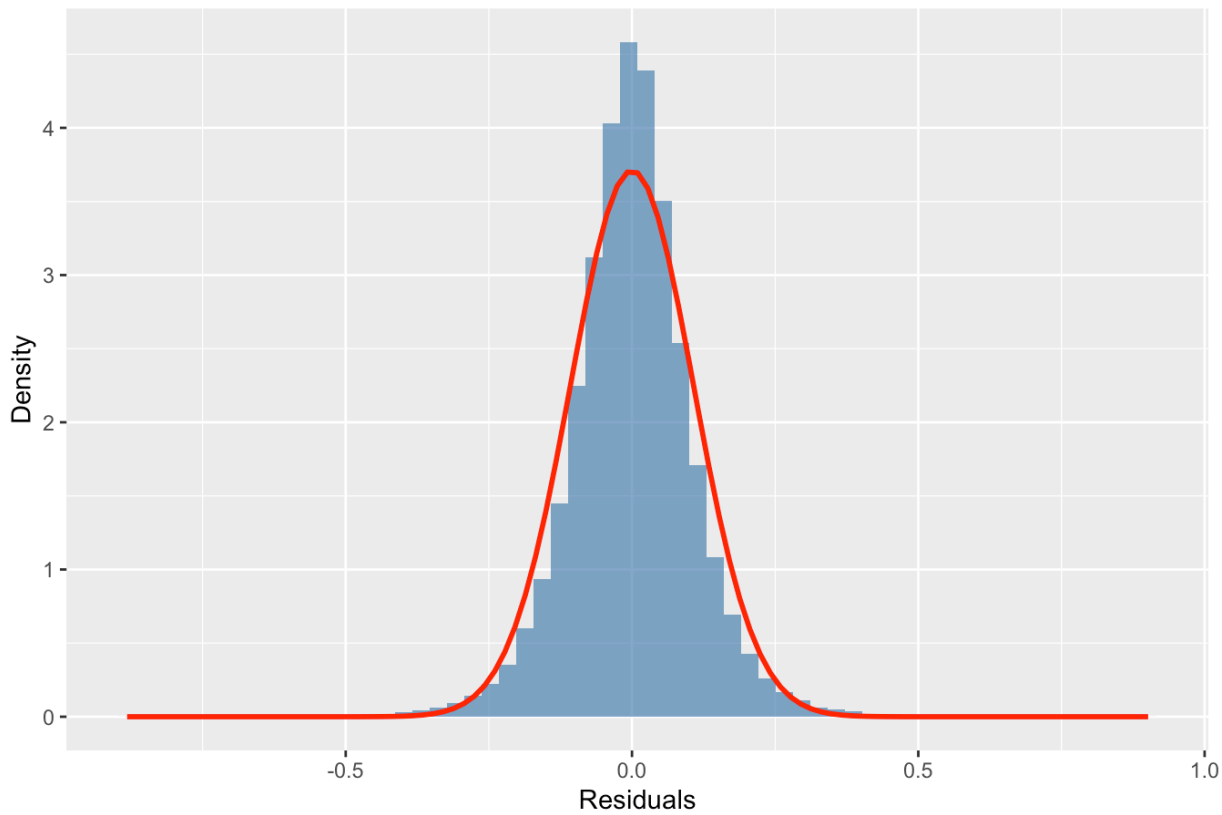
```
##
##  Jarque Bera Test
##
## data:  m4_data$residuals
## X-squared = 32990, df = 2, p-value < 2.2e-16
```

We reject the null of the Jarque-Bera Test ($p\text{-value} < 2.2e-16$) and conclude that residuals are still not normal, but that is to be expected with this many observations. The main issue is heavy tails rather than any particular trend.

Residual Histogram

```
#Histogram of Residuals
ggplot(m4_data, aes(x = residuals)) + geom_histogram(aes(y = after_stat(density)), bins = 60,
  fill = "steelblue", alpha = 0.7) +
  stat_function(fun = dnorm,
    args = list(mean = mean(m4_data$residuals), sd = sd(m4_data$residuals)),
    color = "red", linewidth = 1) +
  labs(title = "Histogram of Residuals", x = "Residuals", y = "Density")
```

Histogram of Residuals



Distribution looks normal, still some outliers and extreme values present with the distribution being taller and narrower than the normal curve with long tails, but overall the residuals are adequate for the sample size.

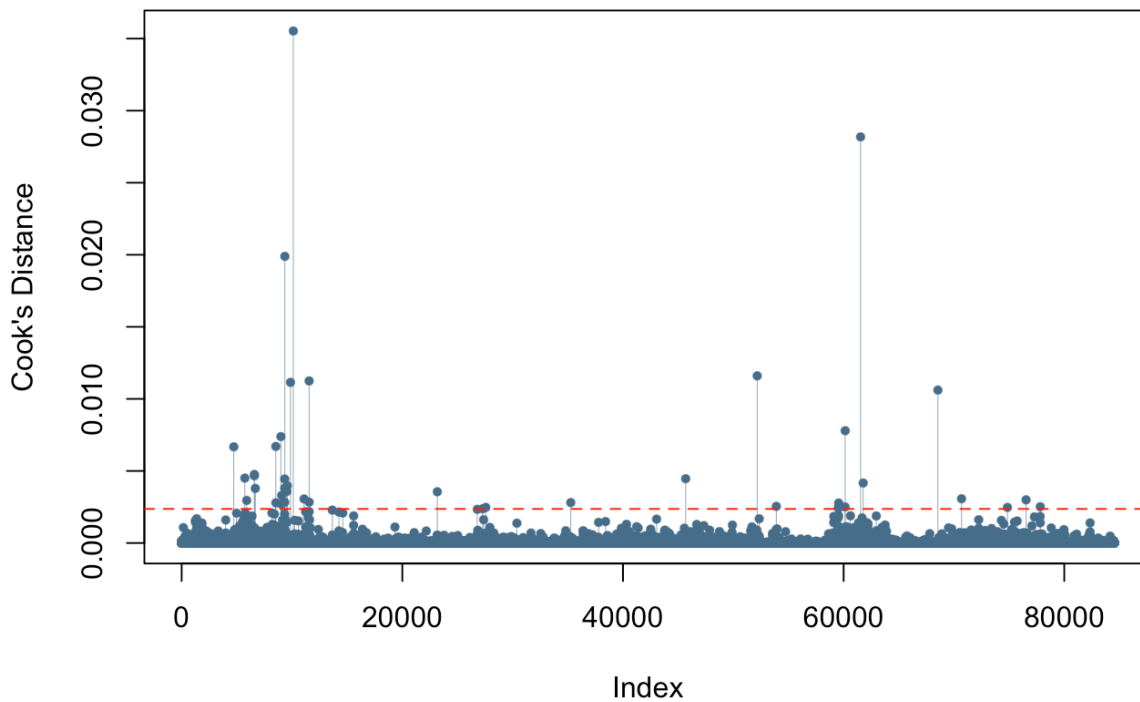
Outliers and Influence

Cook's Distance Influence Analysis

```
# cooks distance for plm
m4_cook <- cooks.distance(m4_lm)

plot(as.numeric(m4_cook), ylab="Cook's Distance", type='o',
     main="Cook's Distance Plot", col="skyblue4", pch=20, lwd=0.25)
abline(h=200/(nrow(dcd.pd)-length(coef(m4_fe))), lty=2, col="red") # to accomodate our large dataset,
we scale up the influential threshold 50x from 4/n-k to 200/n-k
text(17, 0.28, "Potential\nInfluential\nObservations")
```

Cook's Distance Plot



```
m4_influential <- which(m4_cook > 200/(nrow(dcd.pd)-length(coef(m4_fe)))) # selecting the topmost influential observations based on the graph
```

Cook's distance reveals quite a few influential observations remain despite our initial removal of the most extreme outliers. This scaled up threshold selects 41 of the most influential observations.

Leverage Analysis

```
leverage <- influence(m4_lm)$hat

# Add leverage to the original data
dcd.pd$leverage <- leverage

high_lev_threshold <- 3 * length(coef(m4_fe)) / nrow(dcd.pd)

top_lev <- dcd.pd[order(-dcd.pd$leverage), c("district", "year", "outcomegap", "leverage")] |> head(
length(m4_influential))

length(leverage) == nrow(dcd.pd) #dimensions check
```

```
## [1] TRUE
```

```
#influential observations from Cook's
cooks_influential <- dcd.pd[m4_influential, c("district", "year", "outcomegap", "leverage")]

intersect(top_lev$district, cooks_influential$district)
```

```
## [1] "ROSEMEAD ELEMENTARY" "DORCHESTER 04"
## [3] "PEGGS" "ORCHARD ELEMENTARY"
## [5] "MARCUM-ILLINOIS UNION ELEMENTARY" "PATERSON JOINT UNIFIED"
## [7] "CUYAMA JOINT UNIFIED" "RAVENDALE-TERMO ELEMENTARY"
## [9] "MUPU ELEMENTARY" "DEHESA ELEMENTARY"
## [11] "BLOCHMAN UNION ELEMENTARY" "BALMORHEA ISD"
## [13] "WINSHIP-ROBBINS" "TRONA JOINT UNIFIED"
## [15] "NEW JERUSALEM ELEMENTARY"
```

Cross referencing Cook's distance and leverage analysis identifies 15 districts flagged by both metrics, indicating they are unusual in predictor space and potentially influential on model estimates.

Coefficient Stability Plot

```
# Robustness check - excluding influential observations
dcd_robust <- dcd.pd[-m4_influential, ]
m4_robust <- plm(outcomegap ~ log(localptax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minority_c, data = dcd_robust, model = "within", effect = "twoways")

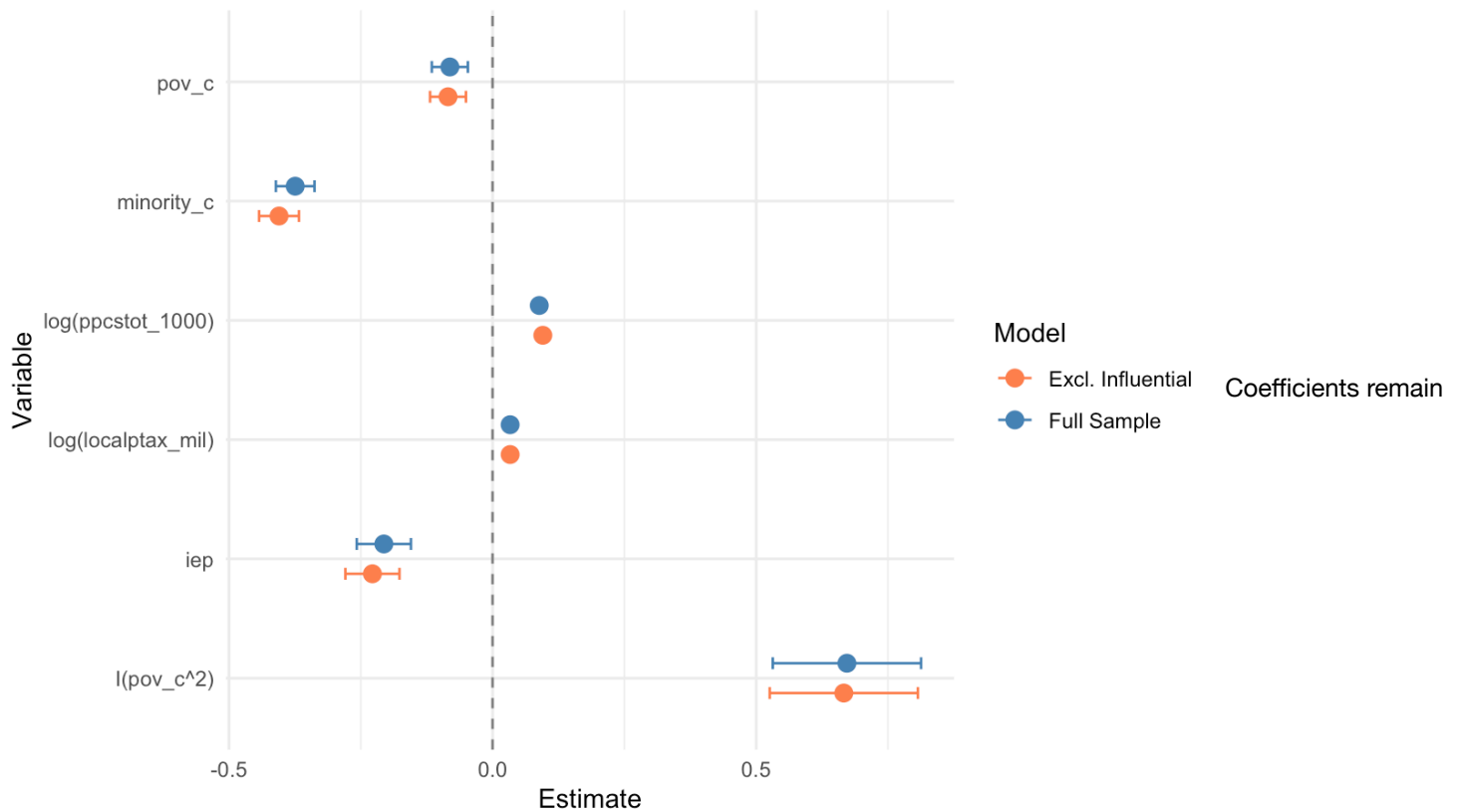
coef_data <- bind_rows(tidy(m4_fe) |> mutate(model = "Full Sample"), tidy(m4_robust) |> mutate(model = "Excl. Influential")) |> filter(term != "(Intercept)") |> mutate(lower = estimate - 1.96 * std.error, upper = estimate + 1.96 * std.error)

ggplot(coef_data, aes(x = estimate, y = term, color = model)) + geom_point(position = position_dodge(0.5), size = 3) + geom_errorbarh(aes(xmin = lower, xmax = upper), position = position_dodge(0.5), height = 0.2) + geom_vline(xintercept = 0, linetype = "dashed", color = "grey50") + scale_color_manual(values = c("Full Sample" = "steelblue", "Excl. Influential" = "coral")) + labs(title = "Coefficient Stability: Full vs Excl. Influential", x = "Estimate", y = "Variable", color = "Model") + theme_minimal()
```

```
## Warning: `geom_errorbarh()` was deprecated in ggplot2 4.0.0.
## i Please use the `orientation` argument of `geom_errorbar()` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
```

```
## `height` was translated to `width`.
```

Coefficient Stability: Full vs Excl. Influential



stable when observations are excluded, influential observations slightly increase the minority coefficient, but largely negligible. All confirmed by the coefficient stability plot, these results support retaining all observations in the final model.

Robustness

Bootstrapping

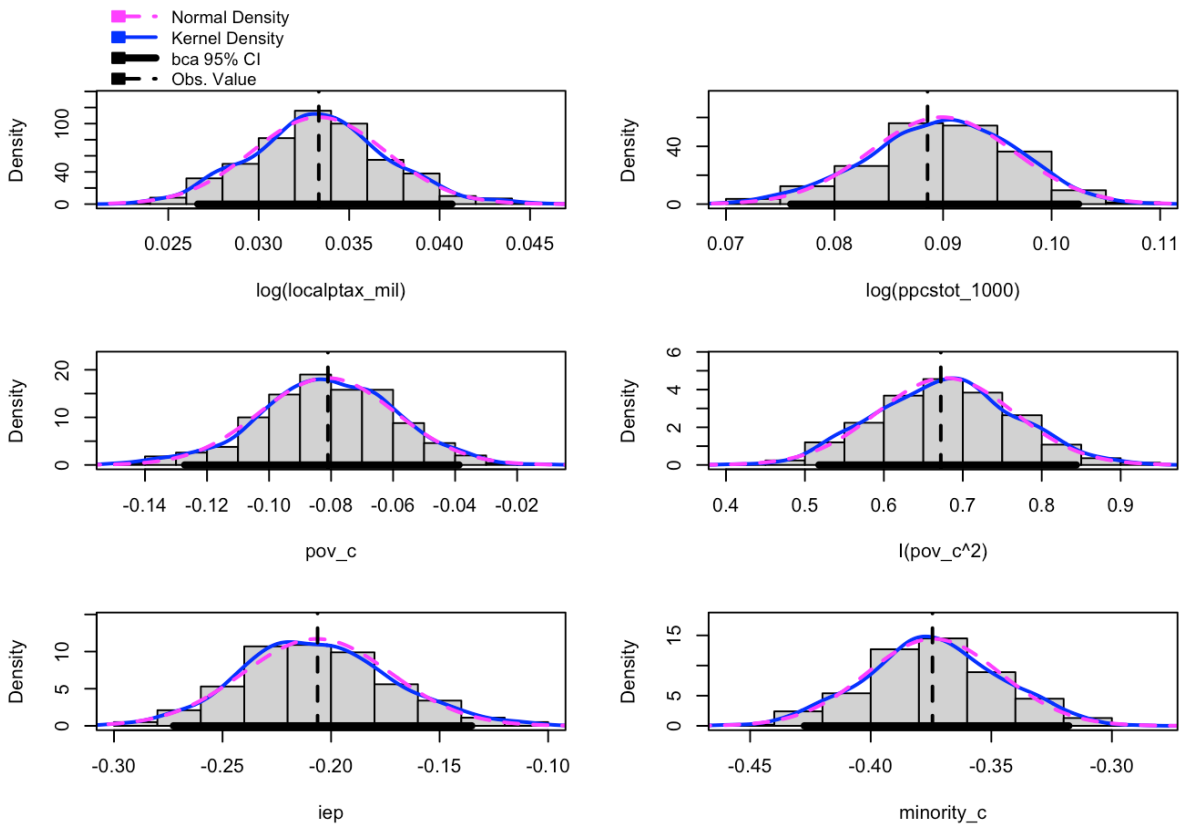
```
m4_fe_boot <- Boot(m4_fe, R=500) # n = 500 given large sample size

summary(m4_fe_boot)
```

```
##
## Number of bootstrap replications R = 500
##
##      original    bootBias    bootSE    bootMed
## log(localptax_mil)  0.033322  1.9766e-06  0.0036943  0.033255
## log(ppcstot_1000)   0.088568  1.3195e-03  0.0066273  0.090196
## pov_c              -0.081016 -1.3431e-04  0.0218452 -0.080850
## I(pov_c^2)          0.672144  6.1366e-03  0.0864857  0.679676
## iep                -0.206217 -3.1387e-04  0.0340824 -0.208038
## minority_c         -0.374396  4.2234e-04  0.0276716 -0.374934
```

Histogram of Bootstrapped Coefficients

```
#Histogram of bootstrapped estimates
hist(m4_fe_boot)
```



Bootstrap validation confirms coefficient stability, with negligible bootBias across all estimates. Bootstrap SE's are consistent with estimates, save `minority_c` where the bootstrap suggests slightly more uncertainty (0.029 vs 0.019). This is confirmed by the histogram, with the 95% confidence interval excluding zero for all predictors. The quadratic poverty term has the widest bootstrap distribution, reflecting greater variability at extremes, but it is positive throughout and therefore remains significant.

K-Fold Cross Validation

```
folds <- groupKFold(dcd.pd$leaid, k = 5)

m4_cv_rmse <- c()

for(i in 1:5){
  # train-test split based on folds
  train <- dcd.pd[folds[[i]], ]
  test  <- dcd.pd[-folds[[i]], ]

  # re-training plm model
  m4_fe_cv <- plm(outcomegap ~ log(localtax_mil) + log(ppcstot_1000) + pov_c + I(pov_c^2) + iep + minority_c, data = train, model = "within", effect = "twoways")

  # testing on hold-out sample
  m4_pred <- predict(m4_fe, test)

  # calculate RMSE
  m4_cv_rmse <- append(m4_cv_rmse, (sqrt(mean((test$outcomegap - m4_pred)^2))))
}

cat("Average CV RMSE ", round(sum(m4_cv_rmse)/5, 5))
```

```
## Average CV RMSE 0.1075
```

Cross validation reveals an average RMSE of 0.1075 indicating predictions of outcome gap vary by 0.1075 standard deviations. This is slightly high considering the scale of the variable but overall a reasonable level of error to expect.

Conclusions

Results and Interpretation

Final Regression Table

```
# final model table
stargazer(m4_fe, type = "text", title = "Final Two-Way Fixed Effects Model",
  dep.var.labels = "Outcome Gap", digits = 3)
```

```
##
## Final Two-Way Fixed Effects Model
## =====
##                               Dependent variable:
##                               -----
##                               Outcome Gap
## -----
## log(localptax_mil)           0.033***
##                               (0.003)
##
## log(ppcstot_1000)           0.089***
##                               (0.005)
##
## pov_c                        -0.081***
##                               (0.017)
##
## I(pov_c2)                    0.672***
##                               (0.072)
##
## iep                          -0.206***
##                               (0.026)
##
## minority_c                   -0.374***
##                               (0.019)
##
## -----
## Observations                 84,531
## R2                           0.012
## Adjusted R2                  -0.123
## F Statistic                  155.133*** (df = 6; 74340)
## =====
## Note:                        *p<0.1; **p<0.05; ***p<0.01
```

Confidence Intervals with Robust standard errors:

```
robust_result <- coeftest(m4_fe, vcov = vcovHC(m4_fe, type = "HC1", cluster = "group"))
robust_ci <- confint(robust_result)
robust_ci
```

##	2.5 %	97.5 %
## log(localptax_mil)	0.02311944	0.04352400
## log(ppcstot_1000)	0.07058230	0.10655438
## pov_c	-0.13943994	-0.02259245
## I(pov_c^2)	0.43650661	0.90778167
## iep	-0.29797301	-0.11446149
## minority_c	-0.44989032	-0.29890107

All coefficients retain their sign, magnitude, and significance under cluster-robust standard errors, with the intervals excluding zero for every predictor. The poverty linear term has the widest interval, reflecting more sampling uncertainty in the within district poverty effect. The minority share is the narrowest, reflecting consistency of this effect.

```
quantile(dcd$pov, 0.75)
```

```
##          75%
## 0.2377964
```

The above table reports the final two-way fixed effects model with cluster-robust standard errors. Coefficients reflect within-district changes over time after removing time-invariant district differences and time shocks.

log(localptax_mil): A 1% increase in local property tax revenue is associated with a 0.00033 standard deviation improvement in the outcome gap, consistent with the theory that better fiscal capacity reduces disparities and improves student test scores.

log(ppcstot_1000): A 1% increase in per-pupil spending is associated with a 0.00089 standard deviation improvement in the outcome gap, exceeding the tax revenue coefficient and suggesting that resource allocation matters more than fiscal capacity alone. This is in line with the theory we cite suggesting that sustained increases in per-pupil spending directly translate to improved test scores and educational outcomes for students.

pov_c and I(pov_c^2): The marginal effect of child poverty rate is $-0.078 + 2(0.657)pov_c$. The turning point of this relationship occurs at a poverty rate of 23.9%, corresponding to approximately the 75th percentile of the sample distribution, meaning that beyond this point additional poverty is associated with improving outcome gaps. This could represent funding mechanisms that disproportionately benefit very high poverty districts. Alternatively, our theory indicates that students in lower-advantaged positions benefit more from sustained funding increases than their advantaged counterparts which may explain some of the effect captured here.

iep: Higher IEP rates are associated with worse outcome gaps; a 1 unit increase in the proportion of students with IEPs worsens the gap by 0.20 units within a district over time.

Minority_c: Most significant coefficient ($t=-20.2187$), a 1 unit increase in the proportion of minority students is associated with a 0.383 unit decrease in the outcome gap variable, meaning disadvantaged students fall further behind.

The within R^2 is 0.012366, indicating that our model explains about 1.2% of the within-district variation in outcome gaps. This is a small percentage but it comes from a model built on almost the entire population of public school districts in the United States with nonetheless highly significant parameters. There are bound to be unique confounding effects that our model could never capture so it is still meaningful to have achieved this level of explanation.

Answering Our Research Question

Question: To what extent does public school funding predict student outcomes when accounting for demographic characteristics and major funding sources?

Our final model indicates that school funding does predict student outcomes and operates through two channels, per-pupil spending ($\beta=0.087$, 95% CI [0.071, 0.107]) and local property tax revenue ($\beta=0.034$, 95% CI [0.023, 0.044]). Per-pupil spending has a stronger effect than local property tax alone, suggesting that actual resource deployment matters more than fiscal capacity. In line with the theory we cited we can see that per-pupil spending increase of only 1% can translate directly into a 0.00089σ change in the gap between a district's test scores and the U.S. average. Likewise, a less direct increase of 1% to local property tax revenue (the primary funding source for most public schools) translates directly into a 0.00033σ change in outcome gaps.

Demographic characteristics play an important role in this relationship, explaining a significant proportion of the variation in district outcome gaps across the country. These effects vary across states and over time consistent with the findings of Allegretto et. al., finding that adequacy varies widely across districts and is lower in higher-poverty districts, many of which also have higher proportions of minority students. The persistence of the minority share effect after controlling for spending suggests that the funding gaps highlighted in the source translate into outcome gaps through ways other than per-pupil expenditure alone. Poverty exhibits a nonlinear relationship with a turning point at approximately the 75th percentile, suggesting that compensatory funding partially offsets disadvantage in the highest quantile districts, this points to outcome disparities reflecting resource distribution beyond aggregate spending levels. Funding matters, but demographics and socioeconomic conditions matter more within districts over time.

Limitations and Extensions

Data coverage is limited by missing outcome gaps in recent years, which can weaken time-trend inference. The analysis also relies on district-level aggregates, so within-district variation (school- or student-level heterogeneity) is not observed.

Funding may be endogenous to outcomes if districts receive resources in response to performance or policy changes, and omitted time-varying reforms could confound estimates. These factors could bias coefficients toward or away from zero. Further, the estimated relationship between funding and outcome gap may differ substantially at the school or classroom level, for example, our outliers found in Cook's distance and coefficient stability are largely smaller district elementary schools. On this point, districts may receive additional state or federal funding in response to poor outcomes, creating reverse causality that biases the spending coefficient downward. The two-way FE partially addresses this issue by absorbing time-invariant factors, but things like state level policy reforms and economic shocks that vary are not fully absorbed.

Future work could instrument for spending using exogenous policy shocks (e.g., state aid formulas), test heterogeneous effects by poverty quartile, and evaluate lagged spending effects over multiple years. Lagged spending could test whether funding takes multiple years to impact outcome.

Results are most applicable to U.S. public school districts in the DCD/EISI panel during the study period; external validity to other countries, private schools, or different policy regimes should be treated cautiously.

References

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